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Chapter 17: Financial Markets as a Clock Room—A Diagnostic from the Fiber Framework

Abstract

The core function of the financial system is to "make clocks"—to convert the operation of the real economy along the time dimension into tradable readings, prices. These clocks have immense social value when they remain anchored in the real. But under certain conditions financial clocks slide from pointing to reality into self-reference. This is a structural tendency that institutions must continuously work against. Taking the institutional evolution of the U.S. financial system as its main line—from the Federal Reserve system established after the Panic of 1907 and the New Deal financial reforms after the Great Depression, to deregulation since the 1980s, the explosive growth of securitization markets and the 2008 financial crisis, and then to post-crisis reconstruction of calibration—this chapter unfolds a diagnosis of finance through the Fiber Framework.

Keywords

financial markets, the clock room, the Fiber Framework, mapping chain, financial crisis, institutional evolution, self-reference, pseudo-calibration

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§0 Analytical Framework of This Chapter

Before entering the institutional-historical narrative, it is necessary to clarify the analytical framework of this chapter. What follows is not a description of financial history, but a diagnosis, through the five-layer progressive structure of the Fiber Framework, of how financial clocks slide from "pointing to reality" toward "self-reference."

The application of the Fiber Framework to financial markets unfolds through five layers:

Layer	Core question	Correspondence in this chapter and later chapters
L0 Mapping-chain structure	How many layers separate the true state from the final reading? What is discarded at each layer? Who controls the mapping function?	This chapter §§1.1-1.2, securitization layered upon itself, chained citation in derivatives, indirect mapping through indexation; Chapter 19, spatialization of the mapping chain—the global 24-hour continuous market and cross-market pricing
L1 Optimization dynamics	What target is each participant optimizing? Is the incentive structure convex?[1] How tightly is it coupled to the true state?	This chapter §3, how institutional evolution changes incentive structure, and §4.2, how the mapping function is rewritten through use; Chapter 18, a China-US comparison of compensation systems
L2 Identification of pseudo-calibration	Which precise numbers conceal enormous uncertainty? How are these numbers institutionalized?	This chapter §4.1, CDO ratings, and §4.2, the procyclicality and pseudo-calibration of value at risk (VaR); Chapter 20, the ideological filter of pricing models
L3 Diagnosis of recursive closure	Are feedback loops functional or pathological? Under what critical conditions does functionality slide into pathology?	This chapter §1.2, partial input networks and recursive components, §5, Minsky dynamics, and §6, the dollar as the recursive loop of the master clock; Chapter 19, the speed predicament and recursive diagnosis of the spatialized mapping chain

[1] In the L1 row above, "whether the incentive structure is convex" refers specifically to the convexity of compensation structure, that is, an incentive structure in which upside gains and downside losses are asymmetric—outperformance above benchmark brings extra reward, while punishment on the downside is limited (Rajan, 2006). This concept runs throughout the finance section and is systematically unfolded in Chapter 18, calibration of compensation systems. It shares a formally similar property with bond convexity, the curvature of the price-yield curve, and option convexity, Gamma, but the domain of application is fundamentally different.

Layer	Core question	Correspondence in this chapter and later chapters
L4 Incentive capture	How does individual optimization produce fiber inflation at the system level? What is the cross-layer transmission chain of incentive capture?	This chapter, end of §1.2 (third characteristic: the self-reinforcing business closed loop of AAA ratings), and §3.2, the weakening of screening by loan originators, conflicts of interest at rating agencies, and higher leverage after the CSE program; Chapter 21, the micro living conditions of typical financial products and front-line practitioners. Note: the analysis of institutional evolution in §3 is **institution-level modulation** across L0-L4, not L4 incentive capture itself

This chapter lays the foundation for the finance section by establishing the core diagnosis and the institutional-historical narrative. Each of the next five chapters, Chapters 18 through 22, develops mainly one of these layers or dimensions, and they are not mutually isolated parallel diagnoses: the structural tension revealed in one chapter often forms the dynamic background against which the next unfolds.

Beyond the five-layer progressive structure, the six chapters of the finance section can also be read along one "drifting from the real into the virtual" cross-layer transmission chain—incentive distortion -> lengthening of the mapping chain -> institutional entropy increase -> ideological masking, four links interlocking end to end into a self-reinforcing loop (for the full formula, the closed-loop logic, and its complementary relation to the five-layer structure—one locates problems, the other explains transmission—see footnote [2]).

In addition to the five-layer progressive structure, the finance section also organizes its six-chapter narrative along three transversal dimensions—time, space, and subject—complementary to the five-layer structure: the time dimension concentrates in this chapter's §5 and Chapter 19, the spatial dimension in this chapter's §6 and Chapter 19, and the subject dimension runs through Chapters 18, 20, and 21 (for the chapter-by-chapter correspondence, see footnote [3]).

[2] **The "drifting from the real into the virtual" cross-layer transmission chain.** Beyond the five-layer progressive structure, the six chapters of the finance section share one cross-layer transmission chain describing the four links of "drifting from the real into the virtual":

[3] **The chapter-by-chapter footholds of the three transversal dimensions across the finance section.** The three transversal dimensions of the finance section—time, space, and subject—correspond across its chapters as follows. **Time dimension**—this chapter §5, the endogenous periodicity of Minsky dynamics; the whole of Chapter 19, the temporal compression of the speed predicament. **Spatial dimension**—this chapter §6, the global transmission of the master clock; the whole of Chapter 19, the spatialization of the mapping chain through cross-border ETFs, 24-hour trading, and the global financial cycle. **Subject dimension**—Chapter 18, compensation and the revolving door: whose incentives? who calibrates?; Chapter 20, the SVO-EMH filter: who prices? whose interests are hidden?; Chapter 21, cases of front-line practitioners and typical financial products: who bears the pressure? These three dimensions and the five-layer progressive structure complement each other: the five layers ask "what do we see at each layer," while the three dimensions ask "along which axis do we run through the whole section."[^4]: **The division of labor between the two analytical frameworks.** The two analytical frameworks in the finance section undertake different tasks. Chapters 17 through 22 use the Fiber Framework for

The finance section uses two analytical frameworks side by side, each with its own task: the Fiber Framework identifies mapping structures, incentive mechanisms, and their feedback loops, and Chapter 22 then introduces the Macro-Cognitive Accounting Framework (MCAF) for cross-period accounting and calibration of completed-cycle cases—the two respectively emphasize mechanism diagnosis and process accounting; they are not substitutes between which one must choose, but complements in a division of labor (for details, see footnote [^fb13_4]).

§1 Why Financial Markets Are the Natural Diagnostic Object of the Fiber Framework

The core diagnostic object of the Fiber Framework established in Chapter 4 is any mapping system from true state to reading. **The true state (T)** is the object the system is trying to capture. **The mapping function (π)** is the conversion rule from the true state to the reading. **The reading (r)** is the output of the mapping function. Given a reading, the set of all true states that can produce that reading is called the **fiber ($\pi^{-1}(r)$)**. The size of the fiber measures the information carried by the reading: the smaller the fiber, the stronger the reading's power to discriminate among true states; the larger the fiber, the more true states are compatible with the same reading, and the lower the diagnostic value of the reading—strong discrimination does not guarantee accuracy, which also depends on whether the mapping itself is valid.

The Fiber Framework was already applied in Part II to artificial intelligence (AI) as a mapping system. Financial markets are the most natural diagnostic object of the Fiber Framework for three reasons.

§1.1 Three Structural Characteristics

First, within the institutional structures of human society, the mapping chain of financial markets has a uniquely strong capacity for multilayer superposition. Layered securitization, chained citation in derivatives, and indirect mapping through indexation can all be superimposed on the same financial product, making the mapping chain of financial markets unusually deep by the standards of other social systems. A precise quantitative comparison across institutions awaits an operational measure of "mapping-chain depth"; at present the framework offers only a qualitative directional judgment. There are at least three mechanisms by which the mapping chain is lengthened.

The first is layer upon layer of securitization. The mapping chain of a single mortgage has only one layer: the borrower's repayment capacity, the true state, -> the loan contract, the reading. Bundle thousands of loans into a mortgage-backed security (MBS), and the mapping chain becomes two layers. Re-slice and recombine the MBS into a collateralized debt obligation (CDO), and it becomes three layers. Bundle slices of CDOs again into CDO^2, and it becomes four layers. At the level of CDO^2, no one can infer from a price the condition of any particular underlying loan.

The second is chained citation in derivatives. The reading of an interest-rate swap does not point to any activity in the real economy, but to another financial reading—the London Interbank Offered Rate (LIBOR) or the Secured Overnight Financing Rate (SOFR). The settlement of a credit default swap (CDS) depends on contractual credit

structural diagnosis, the L0-L4 layers, dissecting mapping-chain structure, incentive distortion, and pseudo-calibration mechanisms. Chapter 22 uses the six variables and three clocks of the MCAF to account for and calibrate completed-cycle financial cases across periods, tracking the accumulation and liquidation of cognitive debt. The division of labor between the two frameworks is not "which one is more applicable," but "which one fits the present analytical task"—structural dissection uses the Fiber Framework; cross-period accounting uses the MCAF.

events, and its price reflects the market's judgment of default probability and recovery—one end of the chain remains tied to the firm's real repayment capacity, but day-to-day pricing relies heavily on references to bond and CDS market quotes. The reading of an option points to the reading of the underlying asset, and that underlying asset may itself already be a derivative. Every additional layer of citation deepens the composite mapping function by one layer; when the new layer merges distinct inputs without adding independent information, the fiber inflates. Bank for International Settlements (BIS) data show that by June 2025 the global notional amount outstanding of over-the-counter (OTC) derivatives reached $846 trillion, up 16% year on year (BIS, 2025a). Notional amounts measure contract size rather than net risk exposure, and interest rate and FX products account for the overwhelming majority—cash flows on such products are linked to other financial readings (reference rates, exchange rates), so the mapping chain extends within the financial system itself. (A note: whether chained citation in derivatives is intrinsically a source of risk is contested in the literature—the Chicago School's Miller (1995) argued that the rapid spread of derivatives occurred precisely because they met genuine commercial needs, and that crashes stem mostly from policy errors rather than from private transactions themselves; this book takes up this symmetric position in Chapter 21 §4, and here takes only the structural fact that "a lengthened mapping chain lets tail risk escape the institutional field of view," without denying derivatives' genuine risk-management function.)

The third is the indirect mapping of indexation and passive investment. When an investor buys an S&P 500 index fund or an exchange-traded fund (ETF) tracking that index, the investor is not judging the true state of 500 firms. Rather, the investor accepts a mapping function determined by the rules of index construction. Market-capitalization weighting means that a rise in the price of large firms naturally increases their weight in the index—a change that in itself requires no additional buying by the fund. Sushko & Turner (2018) point out that the shift toward passive investing could affect securities markets in two key ways: first, higher correlation of returns and less security-specific price information; second, effects on aggregate investment fund flows and market price dynamics.

This migration has already crossed a critical point: Morningstar data show that the net assets of passively managed funds surpassed those of actively managed funds for the first time in 2023, and that by the end of 2025 passive funds held $19.42 trillion in net assets, more than 55% of the U.S. fund market, against $16.02 trillion for active funds (Morningstar, 2026). The picture at the flow level is equally one-sided: in 2025 alone, U.S. equity funds saw roughly $386 billion of net outflows from active products, while passive equity categories took in more than $380 billion (Morningstar, 2026).

Only when new money keeps flowing into passive products and index rebalancing interacts with price shocks does the positive-feedback loop of "passive flows reinforcing the trend in large-cap stocks" form (Sushko & Turner, 2018)—precisely the concrete mechanism by which functional recursion slides into pathological recursion. Passive investment does not lengthen a single mapping chain, but inserts an opaque middle layer into the mapping function: between the investor and the true state of the invested firms stands an automated weighting rule.

The result of the superposition of the three mechanisms is that the depth of the mapping chain in financial markets is unique in institutional history. No other social system allows mapping chains to be superimposed through so many layers while still being treated by participants as objects that are "priceable." It is worth noting that the three mechanisms continue to deepen. The BIS Triennial Survey 2025 shows average daily turnover in global foreign exchange at $9.6 trillion, up 28% from 2022, and average daily turnover in OTC interest-rate

derivatives at $7.9 trillion, up 59% (BIS, 2025c). The mapping chain has not shortened; it is accelerating in length. Philippon's 2015 diagnosis that the unit cost of U.S. financial intermediation had not declined over 130 years, and had actually risen over the last forty years, will be taken up in the context of deregulation in §3.2.

The three mechanisms are not mutually exclusive. They can be superimposed simultaneously on the same financial product. An ETF option superimposes both passive investment, the underlying ETF is managed by automated rules, and chained derivative citation, option -> ETF -> index -> constituent stocks, thereby lengthening the mapping chain at two levels. Their cross-superposition means that the actual length of the mapping chain may be far greater than any single mechanism alone would suggest.

Second, the mapping chain of financial markets is continuously lengthened in the spatial dimension, while the independent component on the input side is systematically diluted. The pricing of each class of financial assets uses as partial inputs the readings of other asset classes, yet retains an anchor in the true state independent of the financial system. Cross-border transmission makes this partial-input network two-way and lengthens it in the spatial dimension. Under extreme stress, the correlation of most risky assets rises sharply and the independent dimension is heavily compressed. At particular nodes, recursive components are superimposed on top of external anchors, but fully closed multi-mapping recursion occurs only under extreme conditions, in Minsky's Ponzi-finance phase and in the death spirals of algorithmic stablecoins[4]. Ordinary financial markets are characterized by the coexistence of the two-waying of partial-input networks and the accumulation of recursive components. By "the two-waying of partial-input networks" is meant that the pricing functions of different asset classes take the readings of other asset classes as partial inputs, and that this input relation is bidirectional at the cross-border level. Yet the input side of each pricing function always retains an anchor in the true state independent of readings internal to the financial system, such as central-bank responses to inflation and growth, or firms' profitability, so the structure does not become a fully closed multi-mapping recursion. This is a fundamental difference between the recursive structure of the financial system and that of AI systems, where models can become trained entirely on the outputs of other models. The technical argument is unfolded in the following subsection.

§1.2 The Two-Waying of Partial-Input Networks and the Accumulation of Recursive Components

> **Note on domain adaptation.** "The two-waying of partial-input networks and the accumulation of recursive components" is not a revision or extension of the core concepts of Chapter 4, but a domain-specific instantiation, in finance, of the degeneration of mapping into self-reference, $\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$. The key difference between the financial system and the AI system—the

[4] **The death spiral of algorithmic stablecoins:** An algorithmic stablecoin (such as UST on the Terra blockchain) does not maintain its dollar peg through off-chain reserve assets (dollars, Treasuries), but through two-way convertibility with another token within the same system (LUNA): the protocol exchanged one unit of UST for one dollar's worth of LUNA, and vice versa, with actual conversion subject to fees and on-chain processing capacity. Liu, Makarov & Schoar (2026) define the mechanism of the "death spiral" as follows: converting UST into LUNA "reduces the UST supply, [but] it also increases the LUNA supply and dilutes existing holders of LUNA"; if this increase in the LUNA supply is expected to lead to a significant decline in the LUNA price, then any LUNA holder would be better off selling LUNA ahead of the conversion resulting in "a so-called 'death spiral' of both UST and LUNA falling in tandem." The study records that the collapse occurred in May 2022 and "wiped out $50 billion in valuation" within three days, and notes that in economic terms UST was "like infinite maturity convertible debt with a face value of $1 backed by LUNA"; unlike stablecoins such as Tether or Circle that are backed by off-chain liquid assets (e.g., treasuries), UST had no off-chain collateral and its value support came entirely from a token internal to the system. It should be noted that the authors do not characterize it directly as a classical Ponzi scheme—the paper explicitly states that "unlike in a classical Ponzi scheme, the amount of subsidies provided by TFL was recorded on the Terra blockchain and, in principle, observable by all investors"—recording only the unsustainable, subsidy-driven dynamic in which newly issued tokens supported existing yields. Uhlig (2022, NBER WP 30256) explains, through holders' heterogeneous exit thresholds, why such crashes unfold gradually rather than instantaneously.

binding of return to risk, and the continuing existence of external anchors—means that in finance the degeneration of mapping into self-reference appears as "the two-waying of partial inputs" in normal markets and the "accumulation of recursive components" in certain asset classes, rather than as the globally homogeneous closure of AI. The validity of this domain adaptation depends on the accuracy of the empirical description in §1.2 (Bruno & Shin, 2015; Ilmanen, 2011; BIS, 2025b), rather than on any incompleteness in the formal system of Chapter 4. The scope of the concept is limited to three kinds of financial phenomena: (a) in panic periods, safe-asset yields move in the direction opposite to real-economic fundamentals, as in the abnormal rise of U.S. Treasuries in 2008; (b) in normal markets, financial products whose pricing models require the readings of other assets as partial inputs, as in calibration chains making an option price depend on the price of a futures contract on the same underlying; (c) self-referentially priced financial products, whose pricing formula contains no term independent of readings internal to the financial system, as in VIX derivatives. The three phenomena differ in their degree of recursive closure—(a) appears only in extreme events, while normal markets remain anchored in external states; (b) is the normal state of financial markets; (c) approaches the limit form of complete closure of financial mapping—but whether closure is high or low depends on whether resources and information capable of constraining pricing and performance still exist outside the loop, not on the order in which the three are listed.

Within a domestic market, the pricing of different asset classes shares a number of common references. Under the capital asset pricing model (CAPM) framework, the expected return of any asset is $E(r_i) = r_f + \beta_i[E(r_m) - r_f]$ (Sharpe, 1964). The risk-free rate (r_f) comes from the bond market, while the market portfolio return (r_m) comes from the equity market. In this framework the two serve as the common pricing references. But this is not closed-loop recursion, because the starting point of the chain, the risk-free rate, is ultimately anchored in the real economy: central-bank interest rates respond to inflation and growth, and inflation and growth are readings of real economic activity. The mapping function of each asset class takes the readings of other asset classes as **partial inputs**, but also contains the risk premium and liquidity conditions specific to that asset class (Ilmanen, 2011). The input is mixed: it contains both readings from other assets and the true state independent of readings inside the financial system.

Cross-border transmission lengthens this chain of partial inputs further. Bruno & Shin (2015) reveal the micro mechanism by which U.S. monetary policy is transmitted to emerging markets through global bank leverage: U.S. monetary easing -> higher leverage of global banks -> expansion of cross-border dollar credit -> rising prices of emerging-market assets. The BIS Annual Economic Report 2025 gives a first-hand diagnosis of the same transmission: "post-GFC structural changes have led to stronger cross-country transmission of financial conditions … this transmission has become increasingly multi-directional. Contrary to conventional wisdom, the United States is not the only major source of international transmission of financial conditions. Instead, US financial conditions are increasingly affected by developments in other advanced economies" (BIS, 2025b). The mapping chain is lengthened in the spatial dimension, while the proportion of the independent component on the input side is diluted. U.S. financial conditions are now themselves partially influenced by overseas readings.

Bruno & Shin (2015; BIS WP 458, 2014) formalize this transmission as a "double-decker model"—local banks borrow dollars from global banks and on-lend to local firms, while global banks finance those cross-border loans through dollar money-market funds in financial centers (Bruno & Shin, 2015). The three-layer dollar chain

implies a pair of structural propositions: "a stronger dollar constitutes a tightening of global financial conditions" and "financial crises are accompanied by dollar shortages." The micro transmission is the risk-taking channel of currency appreciation: appreciation of the local currency -> strengthening of the local borrower's balance sheet -> decline in credit risk -> expansion of bank lending capacity (same source). Their panel evidence across 46 countries finds that both the level and change of bank leverage constitute a "supply-push" factor, closely related to implied volatility in the S&P 500, the VIX. Gourinchas & Obstfeld (2012) independently record the same pattern—from 1973 to 2010, "rapid increases in leverage and large appreciations in the real exchange rate" were the most robust predictors of financial crisis, in both emerging and advanced economies (quoted in Bruno & Shin, 2015). This micro foundation aligns with the diagnosis in §1.2 of the two-waying of partial-input networks: cross-border dollar credit synchronizes local banks in 46 countries through a global factor, the VIX, and the dilution of independent components is not merely the superposition of input readings, but a structural homogenization realized through the resonance of leverage cycles.

Under extreme stress, the partial-input network can collapse into a single factor. Forbes & Warnock (2012), analyzing capital flows across 56 countries, find that extreme capital-flow events are driven primarily by global factors, especially the VIX. Ilmanen (2011) records the same phenomenon from another angle: in normal periods different asset classes have independent drivers of risk premiums and only moderate correlation; in crises the correlation of strategies bearing tail risk suddenly rises sharply (Ilmanen, 2011). He specifically marks the pattern in three crisis waves: safe assets in the government-bond class "diversify best just when diversification is most needed (1998H2, 2001–2002, 2007H2–2008) and other diversifiers fail, they hedge equity markets best when equities are suffering most, and they benefit from 'flight-to-quality' flows during financial crises" (Ilmanen, 2011). The mirror image is that a wide range of strategies bearing tail risk suffered concentrated loss at the same time: "The coincidence of losses could reflect similar fundamental (tail) risk exposures but also liquidity-induced and positioning-induced correlation spikes (the latter were especially apparent in 1998, 2007, and 2008)" (Ilmanen, 2011). The independent dimensions of the mapping functions inside the partial-input network are compressed almost to zero.

Even so, the two-waying of partial-input networks is still not yet multi-mapping recursion. Federal Reserve decisions remain anchored in domestic inflation and employment; emerging-market central banks remain anchored in domestic inflation and output; the external anchor is always present. The truly recursive component emerges at a deeper level. Jiang, Krishnamurthy & Lustig (2021) find that U.S. Treasuries provide a unique "convenience yield"—global investors are willing to accept lower returns in order to hold safe dollar assets. That component of return corresponds to the global demand of investors for safety and liquidity rather than to the repayment state of any single firm.

When demand for safety rises, the convenience yield widens and Treasury yields fall; falling yields then reinforce the belief that "Treasuries are safe assets." Between input and output a closed loop forms. Over the short time scale of panic this loop can reinforce itself: the source of the 2008 subprime crisis was the United States, yet Treasury yields fell rather than rose. By the conventional ordering of asset risk, the sovereign assets of the country where a crisis originates should come under pressure. This anomaly shows that global investors, in panic, were flooding into safe dollar assets—that is the convenience yield at work.

The service provided by safe assets has real economic content; what must be identified separately is whether this demand, reinforced by rising prices, can override the judgment of repayment fundamentals. This is not completely closed recursion—the external anchor, the U.S. government's taxing power and monetary sovereignty,

always remains—but rather **a recursive component superimposed upon an external anchor**.

True closure of multi-mapping recursion occurs only under even more extreme conditions: the Ponzi-finance phase in Minsky (prices supported entirely by expectations of the next buyer), the death spiral of algorithm-based virtual currencies with no redemption channel into underlying assets, or the self-referential pricing of VIX derivatives, where volatility expectation itself becomes the traded object. The common feature of these cases is that within the loop no mapping system directly touches the true state. Cross-asset pricing in ordinary financial markets is the coexistence of the two-waying of partial-input networks and the accumulation of recursive components—the input side contains both the true state and the readings of other assets, and the ratio between the two changes with market conditions.

The two-waying of partial-input networks and the accumulation of recursive components provide the micro foundation for the Minsky dynamics of §5. When the share of independent components, external anchors, is systematically diluted and recursive components come to dominate, the system slides from functional recursion into pathological recursion. The critical condition of that slippage is further unfolded in §5 through Minsky's three financing states.

Third, fiber inflation in financial markets has an endogenous self-reinforcing dynamic. In most mapping systems, fiber inflation is caused by exogenous factors, technical limitation or information asymmetry. When buyers cannot identify the hidden uncertainty and rating symbols still confer investment access and capital-measurement advantages, however, fiber inflation creates profit—the packaged surface appearance of "certainty" that can be sold thereby finds a market. What rating agencies sell is not necessarily a true judgment of credit quality, but a symbol, AAA, that appears precise. Its commercial value comes precisely from hiding the true size of the fiber. CDO issuers pay rating agencies, rating agencies provide AAA, AAA allows pension funds and insurance companies to buy the product, demand expands -> more CDOs are manufactured -> demand for more ratings increases. This is a self-reinforcing commercial loop. Similar incentive structures operate at other layers as well—the relaxation of screening standards at loan origination and the endogeneity of regulatory-capital models on the position side. The micro causal evidence and institutional implications of these two mechanisms are discussed in §4.2. From the perspective of participants, creating and maintaining fiber inflation in financial markets is an incentivized business model; from the perspective of the system, it is structural failure. The two perspectives coexist and cannot cancel one another out.

One key channel of this incentive structure is not packaging and selling or position construction, but the daily maintenance of pricing models through calibration. Derivatives-pricing models need to be repeatedly re-parameterized according to market prices, a process called "calibration." Calibration gives the model a temporarily market-fitting mathematical appearance. From within industry practice, Wilmott & Orrell (2017) reduce the substance of calibration to one sentence: "The real purpose of calibration, it seems, is to fix the appearances of the model, and provide what looks like a mathematically consistent story." The same logic runs through the whole model ecology. The practitioner's primary goal is not to debunk the model—"after all, there is no bonus for that. Instead, it is to justify the use of the calibrated model, to himself, to his boss, to risk managers, regulators, and investors. There are strong incentives to go with the flow" (Wilmott & Orrell, 2017). In the end, "accuracy isn't really the point. What counts is the impression of accuracy, which is much better served by a model in which the economy is at equilibrium, and risk can be precisely calculated, than one in which the economy is far from equilibrium and risk is essentially unquantifiable" (Wilmott & Orrell, 2017). This supplies a fourth channel parallel to the preceding three, not duplicated by external academic criticism

(Danielsson, 2002; Keys et al., 2010)—beyond ratings on the product side, VaR on the position side, and origination fees on the origination side, the daily maintenance of pricing infrastructure itself is endogenously captured by incentives. In fiber language, the daily calibration of models by practitioners is not shrinking the fiber, but systematically maintaining a fiber that is not small enough, yet tacitly accepted as "acceptable" by all sides. The surface appearance of certainty is itself the commodity, and the commercial value of the "precision of the impression" depends precisely on the invisibility of the real fiber.

These three characteristics—the multilayer superposability of the mapping chain, the continuous lengthening and two-waying of partial-input networks in the spatial dimension together with the accumulation of recursive components at particular nodes, and the endogenous incentivization of fiber inflation—make financial markets the most extreme empirical testing ground of the Fiber Framework. The task of this chapter is to unfold a fiber diagnosis within this terrain.

§1.3 The Distinction Between Bidirectional Coupling and Functional Recursion

Before proceeding, one fundamental difference between the financial mapping chain and the AI mapping chain, Part II, must be made explicit. The AI mapping chain discussed in Part II is predominantly one-way dimensional reduction—world -> human experience -> language -> training data -> model output—and its recursion is primarily pathological in form, model collapse and data contamination. The financial mapping chain is **bidirectionally coupled**: true state (T) -> price reading (r) -> cost of capital -> investment decision -> rewriting of the true state. Prices do not merely reflect reality; through the cost-of-capital channel they also shape reality. This difference determines all subsequent analysis. Financial recursion must be divided into **functional recursion**, prices guide resource allocation and this is part of the designed function of the system, and **pathological recursion**, a self-referential closed loop in which prices detach entirely from the true state, Minsky's Ponzi-finance phase. The aim of diagnosis in this chapter and the following five is not to eliminate recursion, which would mean abolishing the function of the financial system, but to identify the conditions under which functional recursion slides into pathological recursion and the institutional designs that can trigger calibration intervention before that slippage occurs.

Bastianello & Fontanier (2024 working paper; the basic model published as "Expectations and Learning from Prices" in *Review of Economic Studies* 92(3), 2025) provide a micro foundation for this bidirectional coupling. They formalize Kindleberger's (1978, quoted via Bastianello & Fontanier) narrative of bubbles as a dynamic theory of **partial equilibrium thinking (PET)**. The core assumption is that "traders fail to realize the general equilibrium consequences of their actions when learning information from prices" (Bastianello & Fontanier, 2024), which triggers a self-sustaining feedback between prices and beliefs: rising prices induce further buying and thereby still further price increases. Borrowing Soros's distinction between "near equilibrium" and "far from equilibrium," the authors characterize this feedback mechanism through two states:

State	Type of shock	Error-correction mechanism	Amplification effect of PET
Near equilibrium (normal times)	Independent and identically distributed shocks	The most informed traders' information advantage remains constant, and "certain corrective mechanisms prevent perceptions and reality from drifting too far apart"	Weak—PET ignores only a small share of the source of price movement
Far from equilibrium (after a displacement shock)	Kindleberger-style displacement shock, railroads, radio and automobiles, the internet, securitization, "something new under the sun"	Even the most informed traders cannot immediately grasp the long-run implications, and the information advantage becomes time-varying	Strong—partial equilibrium thinking "generates much more amplification than in normal times"

This distinction directly supports the core distinction between functional recursion and pathological recursion. Functional recursion, price signals guide capital allocation -> firm investment -> changes in real earning capacity -> price adjustment, corresponds to PET's "near equilibrium," where the amplification effect is constrained by firms' real cash flow. Pathological recursion, price signals cite only other price signals in a self-reinforcing closed loop, corresponds to PET's "far from equilibrium / after a displacement shock," where the amplification effect loses external constraint.

This bidirectional coupling has three special properties that distinguish it from the feedback loop of AI systems, Part II. (1) **Singleness of the transmission medium**—price is a single scalar observed and responded to synchronously by vast numbers of participants. The synchrony of recursion is therefore extremely strong, allowing the financial system to slide from functionality into pathology within a very short time, as on Black Monday 1987 and in the following trading days. (2) **Institutionalized irreversibility**—prices are embedded in legal contracts, futures settlement, mortgage loans, and so on. Once legal consequences are triggered, margin calls, bankruptcy liquidation, they can hardly be reversed. (3) **Suppressibility of the external anchor**—markets can temporarily ignore fundamentals, as in bubble periods, allowing recursion to slide from functional to pathological and display the characteristics of critical collapse rather than the gradual drift found in AI systems. The endogenous mechanism of that critical collapse is taken up in §5 through Minsky dynamics.

§2 The Essence of Finance Is the Making of Clocks

The real economy runs in the time dimension. Farmers sow and harvest across a full growing season. Factories produce along a physical cycle that turns raw materials into products. These processes have physical causal chains—not compressible, not accelerable beyond physical limits.

What finance does is to convert the running of economic activity along the time dimension into tradable readings—clocks. In *Money Changes Everything*, Goetzmann defines financial technology as "a time machine we have built ourselves. It can't move people through time, but it can move their money" (Goetzmann,

2016). Hayek argued that the core function of the price system is this: knowledge of the various particular circumstances of the economy "never exists in concentrated or integrated form but solely as the dispersed bits of incomplete and frequently contradictory knowledge which all the separate individuals possess" (Hayek, 1945). The most important characteristic of the price system is "the economy of knowledge with which it operates": "[i]n abbreviated form, by a kind of symbol, only the most essential information is passed on and passed on only to those concerned." "In a system in which the knowledge of the relevant facts is dispersed among many people, prices can act to coördinate the separate actions of different people" (Hayek, 1945). Financial prices are the unfolding of this mechanism in the time dimension—they compress into tradable numbers the partial judgments of countless people concerning the future. Goetzmann identifies four key elements of finance: reallocating economic value through time, reallocating risk, reallocating capital, and expanding the channels and complexity through which resources are reallocated (Goetzmann, 2016). In the Fiber Framework, the four each correspond to a particular mapping operation: reallocating value through time maps the future true state T, future repayment promises, into a present tradable reading r, the bond price; reallocating risk involves redeploying, among different holders, who bears the various consequences of one true state, insurance, derivatives; reallocating capital transforms a present r into a bet on future T, equity investment; expanding channels increases the coverage range of the mapping function, securitization turning originally untradeable loans into tradeable readings.

Financial instruments are precisely such mapping devices. A bond says, "principal plus interest will be repaid three years from now." The price of a share corresponds to the market's judgment of the discounted value of the company's future profits. A futures contract says, "wheat will be worth this price three months from now." Every financial instrument is a mapping function—it transforms the true state of the real economy (T) into a tradable reading (r).

The "window" of the financial clock is the cash flow of the underlying asset—the sales revenue of a firm's products, the wage income of a borrower, the yield and price of crops. The inputs of the mapping function (π) of a mortgage include the borrower's credit score, proof of income, and property appraisal. Its window is the borrower's repayment capacity, a reference point in direct causal contact with the state of the real economy. When this window exists and is transparent, the financial clock points to the real. But when the mapping chain is superimposed layer upon layer—MBS -> CDO -> CDO^2—and reaches the fourth layer, the concrete information of the underlying assets can no longer be penetrated. The window frame remains, but what lies beyond the window can no longer be seen.

The social value of the financial clock lies in the allocation of resources across time. One side has money today but does not need to use it; the other side needs money today but does not have it. Finance coordinates the two through debt contracts. The "usefulness" of finance does not come from readings themselves, but from the relation between readings and the real. Bond prices tell holders whether the firm can repay principal and interest on time in the future. Stock prices tell the market how the company is expected to perform in the future. These readings have meaning only insofar as they can be traced back to states of the real economy.

But the validity of the Hayekian proposition depends on one premise: participants in the price system possess private information independent of prices. A farmer knows his harvest and his costs, because he takes part in production. A factory owner knows the cost of his product, because he organizes production. When enough participants in a market cease to possess such independent information and instead take other prices as inputs, price discovery degenerates into self-reference.

It should be marked in advance that the "calibration" discussed in the finance section is limited in scope. Calibration can shrink the fiber—the first two layers are mitigable through continuous work—but it cannot eliminate the fiber; the third layer is irreversible. The core diagnosis of this chapter is not "finance can never be calibrated," but rather "the irreversibility of the third layer means perfect calibration is impossible—calibration of the first and second layers requires institutions to do continuous work, and in the absence of institutions the system tends toward self-reference." This is consistent with the position of the AI section, Part II: the fiber is always non-singleton, but the establishment or dismantling of institutions changes the size of the fiber in the first two layers, keeping deviation within a manageable range or allowing it to slide into the unmanageable.

"The essence of finance is the making of clocks" is a descriptive proposition; "finance serves the real economy" is a normative proposition. The tension between the two is precisely the core diagnostic object of the finance section. Under what institutional conditions does the clock point to reality, and under what conditions does it degenerate into self-reference? This chapter answers the first question, the conditions for pointing to reality. The analysis of partial-input networks in §1.2 and Minsky dynamics in §5 answer the second, the dynamics of slippage toward self-reference. The four cases in Chapter 22 answer the third, whether calibration can be reconstructed ex post. The gap between the descriptive proposition and the normative proposition is not a defect in the argument of the finance section; it is the very object of argument in the finance section.

§3 Institutional Evolution of the U.S. Financial System

§3.1 From Pre-Calibration to the Establishment of Calibration Institutions (1792–1940s)

On May 17, 1792, twenty-four brokers signed the Buttonwood Agreement under a buttonwood tree at 68 Wall Street, the origin of what became the New York Stock Exchange. For more than a century thereafter, the regulation of U.S. financial markets remained fragmented between state and federal authorities—federal regulation of national banks had existed since 1863, but securities markets and most banks remained under state law, while a central bank, deposit insurance, and federal securities regulation were all absent—and banking panics erupted repeatedly: 1819, 1837, 1857, 1873, 1893, 1907—on average less than once every twenty years.

The Panic of 1907 was the turning point. The collapse of Knickerbocker Trust triggered runs in succession, and J.P. Morgan organized a private rescue—locking bankers inside his library until they agreed to contribute funds to stabilize the market. The lesson of that crisis was that the financial system could not rely on the goodwill of private bankers to maintain stability. In 1913 the Federal Reserve Act was passed and the Federal Reserve System was established, the first institutionalized lender of last resort in the United States.

This crisis-driven pattern of institutional construction forms the core divergence between the U.S. and Canadian paths: the American cycle of "crisis -> institution -> loosening -> crisis" is itself the macro manifestation of mapping functions repeatedly being rewritten by their users, §4.2, in sharp contrast to Canada's ex ante design of a stable framework through its 1871 Bank Act. The full comparison between the two structures appears in §3.3.

The crash of 1929 exposed a deeper problem. During the bull market of the 1920s, investors could buy stocks on 10% margin, ten times leverage. Brokers lent to customers, banks lent to brokers, and the credit chain was layered upon itself. The Dow Jones fell 89% peak-to-trough in the stock market (Kobrak & Martin, 2022). There was no firewall between banks and securities firms—banks invested depositors' funds in the stock market, and when banks failed depositors lost everything. Between 1929 and 1933, more than 5,000 banks—nearly one-

third—failed in the United States, with 7% of deposits lost, and in 1933 alone, the final year of the crisis, 4,000 banks failed (same source).

Roosevelt's New Deal established a systematic regime of financial regulation. The Glass-Steagall Act of 1933 established a business separation between commercial banking and investment banking—commercial banks could accept deposits and make loans, but not underwrite securities (with exceptions such as government bonds); investment banks could underwrite securities, but not accept deposits. In the same year the Federal Deposit Insurance Corporation (FDIC) was created to insure covered bank deposits, sharply weakening the incentive for deposit runs. In 1934 the Securities and Exchange Commission (SEC) was established, forcing listed companies to disclose financial information regularly. The common logic of these institutions was to limit the scope of financial institutions' business, force informational transparency, and provide a safety net.

From the 1930s to the early 1970s, the United States experienced nearly forty years of relative stability in finance. Banks operated under Regulation Q—deposit rates were capped, profits came from the spread between lending and deposit rates, and the business model was simple and transparent. No systemic banking crisis occurred in that period. Finance really did "serve the real economy"—not because bankers were morally better, but because institutional design limited what finance could do. The costs were real too: rate regulation suppressed returns to depositors, and credit rationing limited financing channels for some borrowers. There is no institution choice without cost. But the stable framework began to unravel in 1971—the closure of the gold window marked the end of the Bretton Woods equilibrium.

§3.2 Loosening of Calibration Institutions and the Lengthening of the Mapping Chain (1971–2008)

In August 1971 the Nixon administration announced the suspension of the dollar's convertibility into gold (the "Nixon shock"), and the Bretton Woods system, anchored on a fixed dollar-gold parity, moved toward dissolution, with major currencies shifting to floating exchange rates in 1973 (Kindleberger & Aliber, 2015). Also in 1973, the Black-Scholes option-pricing formula was published (Black & Scholes, 1973), and the same year the Chicago Board Options Exchange (CBOE) began trading standardized stock options—MacKenzie (2006) recorded how the formula and the exchange were born almost simultaneously and how the formula went on to reshape the options market in turn. In 1971 NASDAQ, the National Association of Securities Dealers Automated Quotations system, went online, the world's first electronic stock market, laying the institutional foundation for the later wave of technology-company listings. The institutional foundations of derivatives markets and electronic trading were built within that decade.

High inflation in the late 1970s changed the equilibrium. When market interest rates persistently exceeded the ceiling on deposit rates set by Regulation Q, depositors shifted their funds out of bank deposits bound by that ceiling and into money-market mutual funds, which were not subject to the ceiling and could offer market-rate returns—this outflow of funds from depository institutions is "disintermediation," which became especially severe after rates surged in 1979 (FDIC, 1997). Mechanistically, this disintermediation was no accident: the "regulatory dialectic" proposed by Kane (1981) reveals its circular logic—political regulation and economic avoidance are forever the two ends of a seesaw, mutually adapting and alternately responding: regulation creates "loopholes" for the avoiders to mine (loophole mining), and avoidance in turn provokes a new round of re-regulation, so that a static equilibrium is nearly impossible (Kane, 1981). Miller (1986) later borrowed exactly this concept to conclude that the principal impulse behind the most successful financial innovations of the past twenty years came precisely from regulation and taxation. The money-market fund is the archetypal product of

this logic: it was itself an instrument born to circumvent the Regulation Q rate ceiling.

More importantly, this thread of disintermediation, opened up by money-market funds, did not later dissipate but grew into an entire "shadow banking" system operating outside traditional bank regulation. Gorton and Metrick (2010) record how the rise of shadow banking benefited from regulatory and legal changes—changes that gave advantages to three types of actors: money-market mutual funds wresting retail deposits from traditional banks, securitization moving traditional banks' assets off the balance sheet, and repo (repurchase agreements) allowing securitized bonds to be "used as money." The money-market funds described earlier in this section and the securitization to be developed below are precisely the first two pillars of this system; and the third pillar—repo financing—ultimately produced in 2008 the "run on repo" demonstrated by Gorton and Metrick (2012): unlike a traditional bank run in which depositors withdraw deposits, this was a systemic run within the "securitized banking" system, triggered by wholesale funders withdrawing repo financing. In other words, a single episode of disintermediation that began in the late 1970s to circumvent rate ceilings ultimately grew into a parallel banking system capable of undergoing a systemic run on its own.

Deregulation also occurred against an anomalous background quantitatively established by Philippon (2015). Defining the "unit cost of financial intermediation" as the ratio of total financial-industry income to the quantity of assets it intermediates, he measures this ratio as stable at about 1.5–2% over 130 years, and finds that it not only failed to decline with advances in information technology, but that the quality-adjusted unit cost, in recent decades, actually rose. The contrast Philippon himself most emphasizes can be seen by comparison with retail (Walmart is the typical case): in IT-intensive industries such as retail and wholesale, technological investment and economies of scale drove unit costs and product prices down continuously, and these industries did not thereby claim a larger share of GDP; finance alone ran the other way—despite enormous IT investment, the unit intermediation cost did not fall, and the share of financial-industry value added in GDP kept rising. The mechanistic picture he offers interlocks directly with this section—after traditional banks were replaced by the "originate-to-distribute" model, intermediation became "a daisy chain," along which the face value of transactions can far exceed the underlying assets, yet from the outside the unit cost actually paid by firms and households did not fall (Philippon, 2015).

As for where this efficiency dividend went, one must separate "the original author's claim" from "commentary based on his research." Philippon himself lists "the dividend being captured by the financial industry itself in the form of higher trading volume and compensation profit" as the most powerful candidate explanation, but cautiously stresses that it remains an unresolved research question, and that his model explicitly does not address financial crises (Philippon, 2015). Industry commentary based on that research pushes the judgment more directly—Wright (2014), in a commentary on the working-paper version of Philippon's article made before its formal publication, notes that the benefits of technology and deregulation were "largely captured by market participants in the form of higher compensation and profits, mainly by greatly increasing trading activity": foreign-exchange trading volume rose about 200-fold since the 1970s, U.S. equity trading rose 20-fold over twenty years, and although unit trading costs fell, the dividend was retained by the financial industry; the same holds in asset management, where the average nominal fee on U.S. mutual funds from 1980 to 2010 did not fall but rose slightly (from 66 to 69 basis points), even though assets under management swelled over the same period from $25 billion to $3.5 trillion and low-cost index funds rose to about three-tenths of the market (Wright, 2014).

The above citations must be distinguished from the judgment of this book. Philippon and Wright answer the

empirical question of "why the efficiency dividend was not transmitted to users"; in the view of this framework, this cost-side fact is directionally consistent with the diagnosis of "lengthening of the mapping chain -> fiber inflation" in §1—it shows that technological investment and the expansion of the chain did not translate into lower unit costs for users; but cost data do not themselves measure the size of the fiber, and the claim that "what deregulation opened was a space for the self-proliferation of the fiber" is this framework's interpretation, not the conclusion of Philippon or Wright themselves.

In 1980 the Depository Institutions Deregulation and Monetary Control Act phased out interest-rate ceilings. Savings and loan institutions (S&Ls) were permitted to undertake a wider range of investments, while deposits remained insured by the FSLIC (Federal Savings and Loan Insurance Corporation). The result: aggressive S&Ls used protected deposits to invest in high-risk projects—commercial real estate, junk bonds, even direct equity investment. By the late 1980s more than 1,000 S&Ls had failed, and the total cleanup cost of the S&L crisis was about $160.1 billion, of which about $132.1 billion was borne by federal taxpayers (FDIC, 1997, *History of the Eighties*). The S&L crisis was the first systemic lesson of the deregulation era: if the business scope of financial institutions is widened without a corresponding expansion of regulatory capability, the risk is ultimately borne by taxpayers.

Securitization developed rapidly in the same period. In 1970 Ginnie Mae provided the first government guarantee for a mortgage-backed security (MBS), the Ginnie Mae I pass-through, and housing-loan securitization began from there. This innovation solved a real problem: local banks had limited funds for lending, and securitization allowed national and even global capital to flow into the housing market, enabling more Americans to buy homes. By the 1990s the market in government-sponsored MBS had become one of the largest bond markets in the world.

In 1998 the collapse of Long-Term Capital Management (LTCM) sent a warning that was ignored. LTCM was managed with the participation of two Nobel laureates, Merton and Scholes, and used complex arbitrage strategies with leverage of about 28:1. As many as fifty-five banks financed LTCM, competing to lower terms in order to secure business with the star fund (Lowenstein, 2000). The Russian default triggered global market turmoil. LTCM began 1998 with $4.7 billion in capital; after the Russian default it lost $1.9 billion in August alone—45% of its capital at the start of that month, leaving only $2.28 billion—and by September, before the rescue, it was nearly exhausted. Federal Reserve Bank of New York President McDonough coordinated a consortium of 14 banks to inject $3.65 billion in order to prevent the fund's over-$1-trillion notional derivatives exposure from generating a systemic chain collapse on default (Lowenstein, 2000). Merton had warned at the time of receiving the Nobel Prize: "It is a misconception to think that because you can measure risk, you can eliminate it." Yet the actual operation of LTCM was built precisely on that misconception. The later fate of the five investment banks in the CSE program, together with the banks that financed LTCM, constitutes two unfoldings of the same pattern. In the same year Brooksley Born, Chair of the Commodity Futures Trading Commission (CFTC), attempted to regulate OTC derivatives, but was jointly opposed by Federal Reserve Chair Greenspan, Treasury Secretary Rubin, and Deputy Secretary Summers, and Congress promptly passed a measure suspending further CFTC action on OTC derivatives (the final legislative closure being the Commodity Futures Modernization Act of 2000 discussed below).

In the following three years, three key institutional lines of defense were dismantled. In 1999 the Gramm-Leach-Bliley Act repealed the core provisions of the Glass-Steagall Act's separation of banking from securities affiliation—financial holding companies could simultaneously conduct deposit-and-lending, securities

underwriting, and derivatives trading within the group (the separation was reconstituted in the form of holding-company subsidiaries, not abolished outright). In the same year Goldman Sachs converted from a partnership into a public company—the era in which partners bore risk with their own money ended, and the era of bearing risk with the money of shareholders and creditors began, and the personal share of risk borne by the firm's executives declined accordingly. In 2000 the Commodity Futures Modernization Act explicitly exempted OTC derivatives, including credit default swaps (CDS), from federal regulation. CDS later became the key channel of transmission in the subprime crisis.

In the early 2000s private-label securitization exploded. Unlike government-guaranteed MBS, private-label products carried no government credit backing; they relied on ratings agencies as the signal of quality. CDO issuance expanded sharply between 2004 and 2007. In 2004 the SEC's Consolidated Supervised Entities (CSE) program allowed the five major investment banks—Goldman Sachs, Morgan Stanley, Merrill Lynch, Lehman Brothers, and Bear Stearns—to calculate capital requirements with approved internal models. Investment-bank leverage had been climbing before and continued to climb after: Goldman Sachs's leverage rose from 17:1 in 2000 to 32:1 in 2007; Morgan Stanley and Lehman rose by about 67% and 22% respectively, both reaching 40:1 by the end of 2007 (FCIC, 2011)—a buildup that straddled the establishment of the CSE program and cannot be attributed wholly to the 2004 change.

In 2005 Rajan, then chief economist of the International Monetary Fund (IMF), presented a paper at the Kansas City Fed's Jackson Hole conference under the title "Has Financial Development Made the World Riskier?" His core argument was that the compensation structure of financial intermediaries systematically incentivizes tail risk—fund managers collect management fees and performance fees in normal years, while losses in disaster years are borne by investors. This asymmetric incentive concentrates the financial system on strategies with a high probability of small positive returns and a low probability of enormous loss (Rajan, 2006). Lawrence Summers publicly criticized Rajan at the conference. Three years later the crisis erupted.

Rajan himself expanded the event systematically in *Fault Lines* (2010). He reduces the original observation that triggered the alarm to the counterintuitive facts visible in the data: "they suggested that banks had become, if anything, more exposed to risk over the past decade" (Rajan, 2010), because "deregulation and developments like securitization had increased competition, which increased the incentives for bankers (and financial managers more generally) to take on more complex forms of risk" (same source). The chapter quotes a 2009 *Wall Street Journal* retrospective, condensing the warning into three mechanisms: compensation asymmetry—"Incentives were horribly skewed in the financial sector, with workers reaping rich rewards for making money but being only lightly penalized for losses … That encouraged financial firms to invest in complex products, with potentially big payoffs, which could on occasion fail spectacularly"; the pseudo-safe appearance of CDS—they were "generating big returns selling these swaps with the appearance of taking on little risk … even though the pain could be immense if defaults actually occurred"; and bank contagion through self-holding of self-created securities—"if those securities ran into trouble, the banking system itself would be at risk … The interbank market could freeze up, and one could well have a full-blown financial crisis" (same source, quoting the *Wall Street Journal* retrospective). Rajan then comments: "Two years later, that's essentially what happened." The first two warnings correspond to the first layer of the three layers of information loss in §4, the invisibility of the mapping function; the third corresponds to the second layer, the rewriting of the mapping function through use—that is, the simultaneous disrepair of the dimensions of "whether there is first-hand penetration" and "whether the mapping is rewritten in the process of penetration" under the four-layer unfolding of the mapping chain in §3.2.

On March 28, 2007, Federal Reserve Chair Bernanke told the Joint Economic Committee of Congress: "At this juncture … the impact on the broader economy and financial markets of the problems in the subprime market seems likely to be contained" (Bernanke, 2007). On the same day Treasury Secretary Paulson assured a House Appropriations subcommittee that "from the standpoint of the overall economy, my bottom line is we're watching it closely but it appears to be contained" (Bookstaber, 2017). Less than three months later, Bear Stearns hedge funds invested in subprime mortgages failed. Between 2007 and 2008, Bear Stearns was sold, Lehman Brothers collapsed, AIG was taken over by the government, Citigroup's senior management had not been notified of $43 billion in concentrated exposure—2% of the company's balance sheet and more than a third of its capital—because it was perceived to be "zero-risk paper" (FCIC, 2011), the Fed cut rates nearly to zero, and quantitative easing began. In September 2008 the SEC terminated the CSE program. Of the five investment banks that had participated, three failed or were acquired, and the other two converted into bank holding companies in order to obtain emergency lending support from the Federal Reserve.

§3.3 Post-Crisis Reconstruction of Calibration and the Global Contrast (2009–Present)

U.S. reconstruction of calibration: FCIC and Dodd-Frank. In 2011 the U.S. Financial Crisis Inquiry Commission (FCIC) released its final report, concluding: "this financial crisis was avoidable. The crisis was the result of human action and inaction, not of Mother Nature or computer models gone haywire" (FCIC, 2011). The report identified widespread regulatory failure, collapse of corporate governance, and excessive risk-taking.

The Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 (Public Law 111-203) was the largest reform of U.S. financial regulation since the New Deal. Its core provisions included the Volcker Rule (Section 619), limiting proprietary trading by banks; mandatory central clearing of OTC derivatives to increase transparency (Title VII); enhanced Federal Reserve supervision and annual stress testing for systemically important financial institutions (Title I); and the creation of the Consumer Financial Protection Bureau to protect financial consumers (Title X) (Dodd-Frank Act, 2010).

LIBOR manipulation and benchmark-rate reform. The LIBOR manipulation scandal of 2012 exposed the structural fragility of the global benchmark interest rate. LIBOR quotes were based not on actual transactions but on the "estimates" of banks—meaning that a handful of traders could manipulate a rate linked to contracts worth trillions globally. Duffie & Stein (2015) systematically analyze the mechanism of manipulation and the path of reform in benchmark rates. Before the scandal exploded, MacKenzie had already recorded the material production process of LIBOR, and with one understated declaration identified that global reading as something assembled from people, equipment, and processes: "The low-key declaration, 'You can publish away', which I heard when standing in the London office in which LIBOR is calculated, initiated its electronic dissemination via multiple data providers. Almost instantaneously, it was possible at more than half a million terminals across the globe …" (MacKenzie, 2009). That material production chain—a few people collecting quotes in a fixed room -> calculation completed at 11:43 -> the instruction "you can publish away" -> simultaneous appearance on more than half a million terminals—is the concrete case of MacKenzie's proposition in the same book that "actors are agencements" (Precept 5): "an actor is 'made up of human bodies but also of prostheses, tools, equipment, technical devices, algorithms, etc.' " (MacKenzie, 2009, quoting Callon 2005). On June 30, 2023, panel-based U.S. dollar LIBOR quotes ceased; the remaining synthetic dollar LIBOR settings served legacy contracts until September 30, 2024, after which SOFR, the Secured Overnight Financing Rate, based on actual transactions, fully replaced them (Federal Reserve, 2023).

China's 2018 Asset Management Regulations. China's New Asset Management Regulations of 2018 represent another path of reconstructing calibration. On April 27, 2018, the People's Bank of China, the China Banking and Insurance Regulatory Commission, the China Securities Regulatory Commission, and the State Administration of Foreign Exchange jointly issued the *Guiding Opinions on Regulating the Asset Management Business of Financial Institutions* (Yinfa [2018] No. 106). The core operations—breaking implicit guarantees, look-through supervision, banning capital pools, and net-value management—are logically analogous to Dodd-Frank: forcing informational transparency and limiting the obscuring role of intermediate layers (PBOC et al., 2018).

The Canadian contrast case. Canada provides a nearby contrast case. The United States and Canada share the longest undefended border in the world, yet their financial systems are radically different. Canada's Bank Act established, beginning in 1871, a tradition of mandatory periodic review, originally every ten years and since 1992 every five years; banking was federally unified; and the six major national banks operated branch networks across the country (Kobrak & Martin, 2022). Kobrak & Martin note that this ten-year review provision is a Canadian institutional particularity. It is a design that endogenizes the risk that the defensive clock (τ_{def}, i.e., the governance clock τ_g) lags behind the projection clock (τ_{proj}, i.e., the techno-capital clock τ_c) by periodically recalibrating the institutional clock.

The cost of the structural difference between the two countries was magnified during the Great Depression: in the United States, "over 5,000 American banks went under, nearly one in three, and 7 per cent of total U.S. deposits were lost"; in Canada, even at the worst point of the Depression, "none went broke. No Canadian banks suffered runs" (Kobrak & Martin, 2022). During the global financial crisis of 2008, no major Canadian bank failed. The IMF assessment before the crisis had already affirmed the resilience of Canada's major banks, a judgment consistent with this outcome: "Canada's financial system is mature, sophisticated, and well-managed. Financial stability is underpinned by sound macroeconomic policies and strong prudential regulation and supervision … The stress tests show that the major banks can withstand sizeable shocks" (IMF, 2008).

Canada is not a perfect contrast case: the asset-backed commercial paper (ABCP) market in Canada did run into severe problems in late 2007, but those problems did not spread into the major banking system (Kobrak & Martin, 2022). Kobrak & Martin summarize the structural difference between the two systems in this way: "Whereas Americans structured a system to achieve maximum access and innovation, Canadians showed more willingness to sacrifice some of both for stability" (Kobrak & Martin, 2022). The cost was real—Canada's financial-market depth and innovative vitality are less than those of the United States—but the outcome in 2008 shows that the return on that sacrifice during crisis was enormous.

The Canadian case demonstrates the existence of a "third option"—allowing financial innovation, but requiring that financial products, at bearable information cost and within regulatory time, be traceable to the underlying cash flows, the bearers of risk, and the resources that service repayment.

FSB international coordination (2025–2026). On November 18-19, 2025, the Financial Stability Board (FSB), at its plenary meeting in Riyadh, formally set out its core work plan for 2026, listing risks from nonbank financial intermediation and stablecoins among its priorities (FSB, 2025, 2026). The meeting explicitly noted that nonbank entities had become major participants in government-bond markets, but that their use of "highly leveraged trading strategies could amplify market volatility and instability, potentially across borders, in the event of a rapid unwinding of positions." It also noted that stablecoins "may improve payment speed and efficiency,

but they raise a number of vulnerabilities, including run risk and regulatory challenges associated with multi-jurisdiction issuers of stablecoins." The FSB's core work plan includes coordinating regulatory frameworks across jurisdictions for crypto-assets and stablecoins, strengthening the resilience of nonbank financial intermediaries, and filling data gaps around leveraged trading in sovereign-bond markets. The governance issue embodied in the Canadian case—how to maintain regulatory coverage and constraints on risk-bearing while allowing financial activity to expand—likewise appears on FSB's international coordination agenda; the two can be juxtaposed within this chapter's framework, but a direct institutional inheritance should not be asserted. Whether the defensive clock (τ_{def}) can catch up with the projection clock (τ_{proj}) remains an open question.

§4 The Three Layers of Structural Information Loss in Financial Clocks

The clock room (Chapter 1, §4) identifies a three-layer progressive irreversibility—information loss, dynamic corrosion, and ontological dimensional reduction—which manifests in this chapter as three kinds of information loss in financial mapping: the invisibility of the mapping function, rewriting through use, and irreversible dimensional reduction (the three propositions of Chapter 4 correspond to these; see §4.1–§4.3). The institutional evolution recorded in §3 can be systematically diagnosed through these three layers. The "three characteristics" of §1 explain why financial markets are suited to fiber diagnosis, the admission condition, whereas the "three layers" of §4 are the unfolding of Chapter 1's three-layer irreversibility in the financial domain; the two do not lie on the same classificatory axis. The three layers are unfolded one by one below—in the 2008 crisis they did not occur sequentially; they were amplified simultaneously, see §4.4.

§4.1 First Layer: The Invisibility of the Mapping Function

Proposition 1 of Chapter 4 states that when the mapping function is not injective, that is, when different true states can generate the same reading, then there exist readings compatible with more than one true state. In financial markets this layer can be unfolded from Akerlof's (1970) general theory, through DeMarzo's (2005) pooling mechanism, Coval et al.'s (2009) amplification through tranching, and Gorton & Pennacchi's (1990) intentional manufacture of information insensitivity, finally reaching Grossman & Stiglitz's (1980) fundamental paradox. This is a progressively deepening chain of argument.

Differences in participants' knowledge of the mapping function create a gradient of detectability, Definition 4 of Chapter 4. Institutional investors may have deep estimates of the mapping functions of a small number of companies, while retail investors know almost nothing. The same price change may be seen by institutional investors as a change in fundamentals, and by retail investors only as numbers jumping on a screen. Akerlof (1970) argued how this form of informational asymmetry changes market composition: when sellers know more than buyers about product quality, market equilibrium can degenerate into one in which trade is dominated by low-quality goods—precisely the mechanism the "lemons market" displays. In his Nobel lecture, Akerlof himself explicitly extended the scope of this proposition to all markets: beginning from large variations in the sales of new cars, he discovered that "the informational problems that exist in the used car market were potentially present to some degree in all markets. In some markets, asymmetric information is fairly easily soluble by repeat sale and by reputation. In other markets, such as insurance markets, credit markets, and the market for labor, asymmetric information between buyers and sellers is not easily soluble and results in serious market breakdowns" (Akerlof, 2001, Nobel Lecture). The CDO market in finance belongs exactly to this composite category of "insurance/credit" in which invisibility of the mapping function falls into the class

Akerlof identifies as "hard to resolve."

The lengthening of the mapping chain multiplies this problem. DeMarzo (2005) proves that pooling itself destroys information: when multiple assets are packaged together and sold, the buyer's sensitivity to information about any single asset is systematically diluted. Coval, Jurek & Stafford (2009) show how slicing and recombination produce deeper destruction of information: senior CDO tranches appear almost default-free, yet their true risk is catastrophic risk—extremely sensitive to assumptions about default correlation in the underlying assets. Once this unobservable parameter departs from the assumption, expected loss can multiply several-fold or even by orders of magnitude. Gorton & Pennacchi (1990) provide the theoretical explanation: one core function of financial intermediation is to create debt that is "informationally insensitive," meaning debt that buyers can hold safely without needing to penetrate the mapping chain. Gorton himself gives a concise definition of this category: informationally insensitive debt is debt that no one need devote a lot of resources to investigating, and it is sold based almost exclusively on its rating; it is designed to avoid exactly that (Gorton, as quoted by Ricks, 2016). Gorton also gave the classical characterization of how crisis erupts: panics occur when informationally insensitive assets suddenly become information sensitive. The holders of such assets do not possess the capability to evaluate them, because the whole point of holding them was "not needing to evaluate them"; once they discover that they must undertake such evaluation, severe problems arise, with system-level consequences (Ricks, 2016 summarizing Gorton's theory). Liquidity evaporates instantly.

Grossman & Stiglitz (1980) formulate the deeper paradox: if prices fully reflect information, no one has an incentive to acquire information; if no one acquires information, prices cannot reflect information. MacKenzie restores the proposition precisely as "markets cannot achieve full informational efficiency … 'there is an equilibrium degree of disequilibrium' is the claim of Grossman and Stiglitz (1980)" (MacKenzie, 2006). Ilmanen gives the same proposition's practical twin from the investment side: "Grossman–Stiglitz (1980) pointed out that in the presence of information-gathering and trading costs, markets cannot be perfectly efficient. Some degree of inefficiency is needed to motivate information gathering and trading" (Ilmanen, 2011). In the CDO market described in §3.2, this paradox appears in extreme form—the cost of penetrating a four-layer mapping chain is exceedingly high, and those who acquire the information may not capture the returns; this weakens the incentive to penetrate and strengthens reliance on ratings—it does not prove that no participant ever undertakes penetration analysis.

Yet the invisibility of the mapping function in finance has one dimension absent from AI systems: **the true state itself is reversely shaped by price signals**. The theoretical model of Bastianello & Fontanier (2024) proves that when traders learn from prices, as they unavoidably do, the bidirectional feedback between prices and beliefs can generate arbitrarily large amplification effects. This means that even if the mapping function were fully transparent, given a price reading r the range of compatible true states T could still be very large—because T itself is formed under the influence of past price signals. What investors observe as "firm profitability" may already have been shaped by historical price signals, financing cost, market sentiment. The fiber arises not only from information loss in the mapping process, but also from endogeneity in the mapped object itself. This is a contextual extension of the definition of fiber in Chapter 4, information loss on the mapping side. In finance, the object being mapped is itself unstable because of recursive components, so the fiber has a dual source, mapping side and object side. But both sources still fit the general form of Chapter 4: the observed reading is not equal to the underlying true state.

The scope of the fiber definition in finance. The formal definition of the fiber in Chapter 4 is

$\pi^{-1}(r) = \{T' \in \mathcal{T} \mid \pi(T') = r\}$—given a reading, the set of compatible true states. The form itself does not presuppose whether the true state is stable. The clock-room thought experiment (Chapter 1) treats the true state at the moment of inquiry as a fixed unknown—the reading lag is handled separately in that chapter's §4.2—and the formalization of Chapter 4 therefore naturally emphasizes information loss on the mapping side. Finance relaxes the presupposition of "stable true state": price signals rewrite the true state through the cost-of-capital channel, §1.3. Fiber inflation therefore has two sources, loss on the mapping side because π is lossy, and drift on the object side because the distribution of T within $\pi^{-1}(r)$ itself shifts over time. Both sources still fit the general form of Chapter 4: observed reading is not equal to underlying true state. The extension here does not contradict the original definition.

§4.2 Second Layer: The Mapping Function Is Rewritten Through Use

Proposition 2 of Chapter 4 states that calibration narrows the identification set by acquiring effective knowledge about the mapping function. Section 4.2 of this chapter examines its mirror structure: the very act of using readings as a basis for decision changes the mapping function itself. The core structure here is **adaptive shift**: when a reading is used as a decision criterion, the mapping function is rewritten under adaptive pressure. This structure repeatedly appears at different layers of financial markets: the firm layer, the market layer, and the model layer.

Goodhart (1984) formalized this dynamic: when a measure becomes a target, it ceases to be a good measure. Lucas (1976) argued the same structure from macroeconomics—once policy rules change, agents' expectations and behavior adjust, and relationships estimated under the old rules cease to hold. In finance, VaR and ratings face exactly the same predicament.

At the firm layer, Graham, Harvey & Rajgopal (2005) find that 78% of chief financial officers (CFOs) admit they would sacrifice long-term value in order to smooth quarterly earnings—the mapping function of earnings per share (EPS) is systematically rewritten by earnings-management behavior. At the rating layer, rating agencies earn their income from the issuers they rate—the mapping function is rewritten by the source of payment. Evidence at the market layer (Keys et al., 2010) and the model layer (Danielsson, 2002) is unfolded in full in the next two paragraphs.

The natural experiment on the FICO 620 threshold by Keys, Mukherjee, Seru & Vig (2010) provides causal evidence for this layer of corrosion. The 620 score itself was not a discontinuity in fundamentals—it was a "rule of thumb" established by Fannie Mae and Freddie Mac and then adopted by subprime securitization investors as an ad hoc threshold (Keys et al., 2010). Precisely because the threshold has no fundamental meaning, the researchers can separate "ease of securitization" from "borrower credit quality." Borrowers with FICO 621 and 619 on the two sides of the threshold have nearly identical credit characteristics, yet loans at 621 are easier to package and sell. The first result is the cliff-like asymmetry in securitization volume across the threshold—"there are more than twice as many loans securitized above the credit threshold at 620+ vs. below the threshold at 620−". The second result is the corresponding reversal in default rates—under securitization, "low documentation loans that are originated above the credit threshold tend to default within two years of origination at a rate 20% higher than the mean default rate of 5%", roughly a one-percentage-point increase. The causal conclusion is explicit: because the only difference between loans on the two sides of the threshold is ease of securitization, the higher default probability of loans above the threshold can only be attributed to

reduced screening by originators. In fiber language, this is **causal evidence that the mapping function is rewritten through use**—borrowers on the two sides of the threshold with nearly identical credit characteristics are treated with lower screening intensity simply because their loans fall on the more easily securitized side; the change on the mapping side comes not from change in the borrowers' true state, but from a change in the incentive structure of "who will bear the default."

At the model layer, Danielsson's (2002) "The Emperor Has No Clothes: Limits to Risk Modelling" provides a precise metaphor structurally isomorphic with Goodhart's law, though pointed in the other direction. He writes: "the statistical process of risk is different from the statistical process of the weather in one important sense: Forecasting the weather does not (yet) change the statistical properties of the weather, but forecasting risk does change the nature of risk" (Danielsson, 2002). On that basis he draws a Goodhart-style corollary: "A risk model breaks down when used for regulatory purposes" (Corollary 1).

This is not an abstract assertion. Danielsson uses the 1998 Russian crisis as the empirical anchor: before the crisis, "most financial institutions employed similar risk model techniques and often similar risk constrains because of regulatory considerations"; when the crisis struck, the volatility of certain assets jumped from 16 to 40, many risk limits were breached simultaneously, and "the presence of VaR based risk limits led to the execution of similar trading strategies, escalating the crisis". His theoretical conclusion is exactly aligned with this evidence: "the distribution of P&L is endogenous to the banks decision-making process. Risk is not the separate exogenous stochastic variable assumed by most risk models; risk modelling affects the distribution of risk". More sharply still: "if regulations restrict the banks scope for pursuing individually optimal strategies, causing banks to act in a more uniform manner during crisis it may lead to an escalation of the crisis".

In fiber language, this is the model-layer version of "the mapping function is rewritten through use" in §4.2—once the same VaR reading is handed to regulation and then synchronously used by multiple institutions, the true risk distribution behind it changes. The randomness of individual behavior is squeezed out by a common model, and previously independent profit-and-loss distributions are coupled by regulation into highly correlated distributions. The statistical properties of the market are rewritten in use. Keys et al. (2010) and Danielsson (2002), from the two ends of micro contract and macro regulation, independently point to the same structure: ordinary use of readings is not neutral observation of the true state, but a reshaping of the true state itself.

MacKenzie (2006) records the same process from the standpoint of financial sociology: financial models do not merely describe markets—they change them. He names the mechanism by which a model, once embedded in market practice, shapes the market in return **Barnesian performativity**, and the mirror concept **counter-performativity**, where use of the model makes the economy become **less like** the model's description of it (MacKenzie, 2006). This is more than the self-fulfilling prophecy of the sociologist Merton (Robert K. Merton, 1948; father of the financial economist Robert C. Merton who appears in the LTCM episode of §3.2). MacKenzie emphasizes that a deeper form of embeddedness is embedding into algorithms, procedures, routines, and material apparatuses. Once so embedded, "An economic model that is incorporated into these can have effects even if those who use them are skeptical of the model's virtues, unaware of its details, or even ignorant of its very existence".

In the case of the Black-Scholes-Merton model, he identifies two historical segments moving in opposite directions. Chicago option data from 1976–1978 show the empirical fit of the model becoming better and better, but that "fit" was itself internal to the process—the model was embedded in practical devices such as arbitrage

spreading, and those practices in turn helped create the very price patterns that the model then described well. In that sense, the performativity of the model was indeed Barnesian.

But after the 1987 crash, "the fit between the Black-Scholes-Merton model and patterns of market prices deteriorated markedly. A 'volatility skew' or 'smile' at odds with Black-Scholes emerged, and it seems to be durable; it has not subsequently diminished or vanished". MacKenzie treats this as candidate evidence of **counterperformativity**: the persistent volatility skew is markedly at odds with the model's predictions, and whether the cause lies in the spread of model-based trading requires further examination of trading mechanisms.

Bräutigam et al. (2023) provide systematic theoretical analysis and numerical evidence for the procyclicality of VaR. Their core finding is that VaR estimation based on historical data is intrinsically procyclical: it underestimates future risk in tranquil periods and overestimates it in crises. More importantly, even when returns are independently and identically distributed, so that no volatility clustering is present, this procyclicality still exists, and the generalized autoregressive conditional heteroskedasticity (GARCH)-type volatility clustering found in real markets further reinforces it. In fiber language, this means that the pseudo-calibration of VaR is not merely a matter of "improper use," assumptions being wrong; rather, even when the assumptions all hold, historical-data-based VaR itself is inherently procyclical. This is a deeper mechanism of institutional lock-in—the statistical properties of the risk-measurement tool are themselves procyclical, providing a tool-level explanation for the rewriting of the mapping by use.

Pop & Pop (2025), in their systematic analysis of Basel III output floors, confirm the incentive of large banks under the internal-model paradigm "to optimize the grading system in order to lower as much as possible the capital requirement". This is a precise instance of regulatory arbitrage. The core function of the output floor is exactly "to limit the capital savings enjoyed by large banks through regulatory arbitrage under the internal-model paradigm." In fiber language, banks are optimizing the reading of regulatory capital, not the true capacity to bear risk. This is the exact institutional manifestation of the second-layer predicament—the mapping function is rewritten through use.

The institutional changes of §3.2 amplified this layer of corruption. Under the originate-and-hold model, loan originators had an incentive to evaluate borrowers seriously, because they themselves retained the risk. Once a securitization market was developed enough that originators could sell loans on quickly and transfer the risk, the mapping function shifted from "assessing true repayment capacity" to "meeting conditions for sale." Change in institutional design altered the speed and degree with which the mapping function became corrupted.

§4.3 Third Layer: Irreversible Dimensional Reduction of High-Dimensional Reality

Proposition 3 of Chapter 4 states that when a mapping sends multiple true states to the same reading, the true state cannot be guaranteed to be recovered from the reading alone—projection from high dimension to low is its typical form. This is the deepest of the three layers—the first two can be continuously mitigated through disclosure, auditing, and incentive constraints, whereas the third cannot mathematically be eliminated. The true state of a company—profitability, technological barriers, management quality, organizational culture, supply-chain resilience—lies in a highly multidimensional space, while the stock price is a scalar. Roll (1988) finds that less than forty percent of the monthly return volatility of the typical stock can be explained by systematic factors—the remaining more than sixty percent is "idiosyncratic volatility." But how much of that idiosyncratic volatility is genuine firm-specific information, and how much is noise? Shiller (1981) finds that stock-price

volatility is far greater than can be justified by subsequent changes in dividends—the "excess volatility puzzle." The two works jointly show that current fundamental frameworks can explain only part of the variation in stock prices—consistent with the direction of irreversible dimensional reduction, though the structural character of the loss in reduction comes from the mapping argument of Chapter 4 rather than from these two empirical findings themselves.

The fundamental nature of this irreversibility is this: even if the first two layers were fully solved, the mapping function completely known and no longer corrupted, many-to-one dimensional reduction—such as the mapping of a firm's high-dimensional true state into a single stock price—would still cause irrecoverable information loss. This means that the goal of financial governance is not to eliminate uncertainty—which is mathematically impossible—but to keep it within a manageable range and to mark its existence honestly.

§4.4 The Superposition of the Three Layers in Crisis

The 2008 crisis displayed simultaneous activation of all three layers. Brunnermeier's (2009) analysis of the 2007–2008 crisis reveals the interaction of two positive-feedback loops: the liquidity spiral, falling asset prices -> forced sales -> further price declines, and the margin spiral, falling prices -> higher margin requirements -> forced deleveraging -> further price declines.

Bookstaber borrows Perrow's theory of "normal accidents" (Perrow, 1984) to provide the structural explanation: when a system simultaneously possesses interactive complexity, components affect one another in nonlinear, unpredictable ways, and tight coupling, failure at one link is transmitted immediately to the next with no buffering time, then, by the core claim of this theory, the risk of systemic accidents is endemic to the system structure rather than merely triggered by external chance. Bookstaber later, in the preface added one year after the book's publication, named this combination "the twin demons of market complexity and leverage-induced tight coupling," and in his 2017 book he cited and systematized the same diagnosis, pointing to complexity and tight coupling as the key components in the origin of market crises. When complexity and tight coupling operate simultaneously on the time dimension, processes advance faster than analysis and response, and Chernobyl- and Piper-Alpha-type disasters are better understood as the realization of structural risk rather than pure accident.

The 2008 financial system met both conditions simultaneously: the interactive complexity among CDOs, CDS, repo markets, and rating agencies, coupled tightly to overnight funding and margin mechanisms. The danger lay not only in any external shock but in the structure of the system that amplified it. All three layers were amplified together in crisis: underlying asset quality deteriorated but the mapping chain was too long for anyone to penetrate, first layer; forced sales rewrote the meaning of price, second layer; and the entire market was compressed into a single liquidity indicator, third layer.

Brunnermeier, together with Crockett, Goodhart, Persaud, and Shin, added three structural conditions to this interaction of the two spirals in the *Geneva Reports on the World Economy 11: The Fundamental Principles of Financial Regulation* published in the same year by CEPR. The report is by the same authors and in large part a parallel translation of the 2009 JEP analysis. First comes a precise statement of the trigger condition: "problems only arise if both funding liquidity dries up (high margins/haircuts, restrained lending) and market liquidity evaporates (fire sale discounts)" (Brunnermeier et al., 2009, *Geneva Reports 11*). The five authors characterize the simultaneous activation of the "two spirals" exactly as this "simultaneous evaporation." Second comes **procyclical leverage**: the self-amplification is not an exception in abnormal periods, but a structural normality.

In the upswing, credit expansion and rising collateral values feed back into one another; in downturns, "the mechanism works in reverse" (same source), and the same structure completes itself in the opposite direction. Third comes a further externality parallel to the two spirals—"credit restriction weakens the economy, which leads to more default and asset price declines, which causes yet more credit restrictions" (same source). This extends the feedback loop from the financial system into the real economy. Correspondingly, the diagnosis of the superposition of three layers in §4.4 means that the condition for simultaneous activation of the two spirals, simultaneous evaporation of funding liquidity and market liquidity, is precisely the compression in time of the second layer, mapping rewritten by use, and the third layer, dimensional loss, into the same moment.

Gennaioli, Shleifer & Vishny (2012) explain why this superposition goes unnoticed before crisis: financial innovation tends to create securities perceived as "safe," but in fact concentrating neglected tail risk. In their 2018 book, Gennaioli & Shleifer systematically expand the 2012 JFE model into an entire chapter specifically discussing the 2008 crisis, and explicitly cite their own 2012 article in a footnote: "The analysis in this chapter is based on two papers by Gennaioli, Shleifer, and Vishny (hereafter GSV 2012, 2013), although here we use a different formulation" (Gennaioli & Shleifer, 2018).

They condense the model into two elements and their consequence: "This chapter presents a basic model of financial fragility with neglected downside risk … The model has two ingredients: strong demand for safe assets and neglect of downside risk." Once the two coexist, "debt is perceived to be safer than it is. In this case, the financial sector expands too much, risk sharing is no longer efficient, and the system is fragile. If bad news brings neglected risks to mind, risk-averse investors dump safe debt on the market" (same source).

More critically, they emphasize that this mechanism does not depend on any assumption of fraud: "there is no trickery by intermediaries in this model. Rather, neglect of risk creates an opportunity for financial innovation, and competitive intermediaries exploit this opportunity to the hilt … More generally, we expect new financial products to load up precisely on the risks investors do not fully appreciate. Under rational expectations, financial innovation would improve risk-sharing and welfare. With neglected risk, in contrast, it creates inefficient risk-sharing and the possibility of crises" (same source).

Investors are not ignorant of the existence of risk; they systematically underestimate the possibility of low-probability events. As §3.2 showed, the mapping chain was in the same period lengthened to four layers, so that no participant whose penetration cost was finite could correct this systemic underestimation. Market forces exploited it to the full.

§5 Minsky and the Directionality of Fiber Inflation

Hyman Minsky (Minsky, 1986) described the dynamics by which finance moves from anchoring in reality to self-reference. **Hedge finance**: the borrower repays both principal and interest out of operating cash flow, and the debt contract remains anchored in the real economy. **Speculative finance**: the borrower can pay interest but must refinance principal, and performance of the contract begins to depend on financial-market conditions. **Ponzi finance**: operating cash flow does not even cover interest, and the unit must rely on new borrowing or asset sales to keep going; performance of the contract depends entirely on financial-market conditions. Connecting this to the recursive vocabulary already established in §1: **hedge finance** is the accounting form under the dominance of functional recursion; **speculative finance** lies in the slippage band from functional toward pathological; and **Ponzi finance** is the accounting form under the full closure of pathological recursion. Ac-

counting form is strongly correlated with recursive form but not strictly synonymous with it—this distinction will be unfolded in empirical form through the four-case calibration of Chapter 22.

Minsky's core proposition is that in stable periods the financial system moves **endogenously** from hedge finance toward Ponzi finance. No external shock is needed, and no one need be malicious. Kindleberger systematized this dynamic in *Manias, Panics, and Crashes*, and Aliber expanded it in the seventh edition to the international dimension and the 2008 global crisis.

Kindleberger & Aliber (2015, Traditional Chinese translation) explicitly list the full sequence of four waves of crisis: the first wave of banking crisis appeared in the early 1980s, when Mexico, Brazil, Argentina, and ten other developing countries together held external debt exceeding $800 billion; the second occurred in the early 1990s, sweeping through Japan and the two Nordic countries, Finland and Sweden; the Asian financial crisis beginning in mid-1997 was the third wave; the fourth began in September 2008, when the two large U.S. government-sponsored mortgage entities Fannie Mae and Freddie Mac were placed under conservatorship, followed within days by the bankruptcy of Lehman Brothers. The seventh edition's causal diagnosis adds that before every wave of banking or financial crisis there was large-scale cross-border capital inflow, rising securities and property prices, and in the years before crisis both external debt and domestic debt grew at rates two to three times faster than interest rates. All four waves followed the same structure: capital inflow -> rising securities and property prices -> debt growing two to three times faster than interest rates -> slowing of inflows -> collapse (Kindleberger & Aliber, 2015).

Galbraith, in *A Short History of Financial Euphoria*, reduces the regularity of financial crisis and forgetting to two structural causes, the first of which directly supports Minsky's proposition of endogenous slippage: "The first is the extreme brevity of the financial memory." "In consequence, financial disaster is quickly forgotten. In further consequence, when the same or closely similar circumstances occur again, sometimes in only a few years, they are hailed by a new, often youthful, and always supremely self-confident generation as a brilliantly innovative discovery in the financial and larger economic world" (Galbraith, 1990). He condenses this into the book's core maxim: "financial genius is before the fall." Financial geniuses are punished when speculative bubbles collapse, yet within a short and measurable cycle another batch of geniuses reappears and financial innovation is again celebrated.

The institutional history of the United States in §3 supplies empirical support to the propositions of Minsky and Kindleberger. The stable framework established by the New Deal ran for forty years; the fading of crisis memory weakened the political will to maintain regulation, and this combined with the inflation, disintermediation, and business-competition pressures described in §3.2—pressure for deregulation was most easily introduced when the system was most stable. Less than ten years after the S&L crisis came LTCM; less than ten years after LTCM came the subprime crisis. Every reconstruction of calibration institutions faces pressure toward dismantlement after a period of stability, because in stable times the costs of calibration are visible, compliance costs and loss of efficiency, while the benefits of calibration are invisible, the crisis that never happened.

Recent work provides operable empirical grounding for Minsky's classification. De Jesus (2025), studying non-listed manufacturing firms in Colombia, finds that deterioration in the interest-coverage ratio is the main determinant of slippage into a Ponzi state, while dependence on imported capital forms a specific channel through which currency shocks are transmitted. More importantly, the study discovers a temporal asymmetry: "contemporaneous sectoral expansions ameliorate immediate default risk through the revenue channel, whereas

lagged growth accumulation is associated with the endogenous generation of future fragility." In fiber language, this is a micro-level empirical corroboration of Minsky's core proposition: stability breeds instability. In the short run growth reduces risk; in the long run it accumulates risk. The time gap between the two is precisely the operating space of the debt clock (τ_d). In this chapter the debt clock remains a qualitative concept; its operationalization will be unfolded in Chapter 22 through ex post calibration of its principal-evidence cases (Japan in the 1990s, the United States in 2008, and China's new asset-management regulations).

Using micro data on 14 listed Chinese banks, Fang Yi & Chen Min (2019) verify the applicability of the Minsky hypothesis to China: low volatility during periods of economic expansion leads banks to take on excessive risk and aggravates the accumulation of systemic risk; the length of China's financial cycle is about 11 years, with medium-run frequency. The key finding is that economic expansion and contraction have asymmetric effects on bank risk-taking: low volatility in expansion is precisely the breeding ground of future fragility (Fang & Chen, 2019). This forms a cross-national cross-check with De Jesus's (2025) finding that current expansion reduces immediate risk while lagged growth accumulates future fragility. Jiang Hai, Zhang Xiaolin, Tang Shenfeng & Chen Chuanglian (2021) further provide the micro transmission mechanism: loose monetary policy significantly aggravates bank risk-taking through the liquidity channel, and the effect is amplified when bank liquidity is higher—"the constraining effect of contractionary monetary policy on bank risk-taking is significantly weaker than the encouraging effect of expansionary monetary policy" (Jiang et al., 2021, author's translation). The two studies together point to the Chinese version of the Minsky proposition: loose environment -> abundant liquidity -> higher bank risk-taking -> accumulation of systemic risk; stability endogenously breeds instability.

This is exactly why the normative proposition of §2—"finance serves the real economy"—cannot be a one-time policy declaration. It is an operation that must be continuously performed. Once the work stops—deregulation, allowing the mapping chain to lengthen without limit, tacitly permitting self-referential loops to form—the system automatically slides toward fiber inflation. This problem cannot be "solved once and for all," just as entropy increase cannot be solved once and for all.

§6 The Master Clock of Global Financial Markets: The Dollar as the Anchor of Recursion

§6.1 The Concept, Historical Forms, and Boundary Samples of the Master Clock

The positioning of the master-clock concept. This section returns to the L3 layer of the five-layer progressive structure of §0, the diagnosis of recursive closure. In the Fiber Framework, "master clock" is an analytical concept, not a permanent assertion about the dollar's status. Within the conceptual system of the book, the master clock is a cross-domain structural concept, fully argued in the finance part, with a corresponding analysis in Chapter 13 §5.5 of the AI part. The master-clock formula—the degeneration of mapping into self-reference ($\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$)—describes the degeneration in which the readings of other clocks substitute for the true state as the input to the mapping function, manifested at the institutional scale of the monetary system. The corresponding forms of this concept across different domains are not fully isomorphic: the corresponding analysis in AI systems is detailed in Chapter 13, §5.5; here no parallel argument is unfolded, and only the definition necessary for the financial scenario is retained.

The Fiber-Framework definition and identifying criteria of the master clock. In the Fiber Framework, when multiple mapping systems coexist and cite one another, a "master clock" may emerge—a clock whose readings become the benchmark against which other clocks define themselves. The master clock is not a necessary

structural outcome, historical monetary systems were multipolar in parallel for long periods, but an emergent result under certain conditions. When a unipolar master clock does exist, it can be identified through three empirical criteria: (1) whether there exists a transmission effect independent of spreads and trade channels; (2) whether there exists a self-reinforcing feedback loop with global financial risk; and (3) whether there are locking network effects. In the current global financial system, the dollar simultaneously satisfies all three criteria. This is the historical product of the institutionalization and reinforcement of dollar centrality under Bretton Woods in 1944 (dollar influence had been accumulating through the first half of the twentieth century, but the single global monetary anchor dates institutionally from 1944), not an inherent property of monetary systems as such.

Historical forms of the master clock: from multipolarity to unipolarity. The global monetary system was not unipolar from the beginning. From the sixteenth century to the middle of the nineteenth, international trade in practice operated in a multipolar structure in which gold and silver standards coexisted. Europe gradually shifted from silver to gold, Britain legally adopting the gold standard in 1816. China, by contrast, from the mid-Ming period, the 1430s, used silver as its main monetary metal, and the Single Whip Reform of 1581 commuted taxes and corvée broadly into silver payments, further establishing silver's central position in fiscal and large-value payments; the system continued for nearly four centuries until the fiat-currency reform of 1935.

Flynn & Giráldez (2002) distinguish two great global waves of silver circulation—the "Potosí/Japan cycle" from the 1540s to the 1640s and the "Mexico cycle" in the first half of the eighteenth century. In the two cycles a large share of the world's silver flowed into China—according to their survey, "from 1571 to 1700, at least half of the silver amounting to 80% of global output flowed into China" (quoting Gu Xiuting, in Guo Hua & He Shijun, 2023, author's translation). Jacks, Yan & Zhao (2017) point out that "[China] was the only large country in the world on a silver standard throughout this period. … While silver was 'just another commodity' in other countries, it was the sole basis of currency in China" (Jacks et al., 2017). Their empirical measurement of silver points in the Shanghai market between 1905 and 1933 shows that the degree of Chinese financial integration was comparable to cross-Atlantic gold points in the same period.

This means that in the so-called classical gold-standard period, the 1870s to 1914, the global monetary system in fact consisted of **parallel gold and silver tracks**, not a single pole: the gold-standard club, Europe, the United States, Japan, and silver-standard China formed two parallel monetary poles, linked through fluctuations in the silver price and trade settlement but not unified under one anchor. The truly global unipolar master clock—one center to which all participating currencies were uniformly anchored—was a product only of Bretton Woods after 1944. Before that, all "master clocks" were regional, their scope limited by the physical networks of precious-metal circulation.

Cross-domain differences in the robustness of the external anchor. The financial master clock is structurally isomorphic to the frontier-benchmark citation form in AI described in Chapter 13, §5.5—one mapping node serves as the reference benchmark for all other nodes—but the two differ structurally in the verifiability of their external anchors. The external anchor of the financial master clock is real-economy cash flow, which has an independent operating time scale that does not depend on financial valuation: during the Great Depression of 1929–1933, variables such as real productive capacity, capacity utilization, and employment continued to operate at the physical and organizational level, providing a reference for the regression of prices and valuations independent of the financial system's own pricing; even when financial valuation deviated for a long time, the operation on the real side continued somewhere to constitute the ultimate source of calibration pressure.

The external anchor of the AI master clock (the γ layer) is the verifiable relation between the output of the evaluated system and the "true answer"—but as the capability of the evaluated system rises, the space of judgments that can be checked at low cost by non-expert humans correspondingly contracts—in the wording of Chapter 13, §5.5, whether this contraction appears, and at what speed, is an empirical question.

The implication of this difference for diagnosis is that when the financial master clock slides from functional recursion toward pathological recursion, there exists at least one independently operating external time scale to serve as a source of calibration pressure—the arrival of that pressure may be delayed (mechanisms such as quantitative easing (QE) and zombie banks can suppress its manifestation for a considerable period) but does not disappear; the AI master clock, by contrast, faces a different diagnostic difficulty—the robustness of the external calibration pressure itself degrades as the capability of the diagnosed system rises, a problem for which the finance section offers no precedent to draw upon; Chapter 13, §5.5 marks "the robustness of the external calibration pressure degrades as system capability rises" as this counterpart difficulty, and the corresponding argument is found in that section.

Case box: Terra-Luna (May 2022)—the failure of an endogenous redemption mechanism with limited external reserves

The discussion above distinguished the financial master clock from the AI master clock in terms of the robustness of their external anchors. Algorithmic stablecoins offer a rare boundary sample: their redemption mechanism is algorithmic at the core, their on-chain data are highly observable, and their off-chain reserves bear no direct redemption function—a combination almost impossible to find in traditional financial data. UST on the Terra blockchain was the largest such case: its core redemption mechanism relied not on redemption out of off-chain reserves but on two-way convertibility with the in-system token LUNA (for the mechanism, see the earlier footnote 5); LFG had built an external reserve of roughly 80,000 BTC before the crash and intervened with it, but the reserve was too small relative to Anchor's high-yield liabilities and the death-spiral dynamics, and failed to prevent the collapse. According to Liu, Makarov & Schoar (2026), on May 7, 2022 two large addresses were the first to withdraw $375 million UST from Anchor, other large holders followed, and Terra "collapsed in three days … and wiped out $50 billion in valuation." The two studies around this event—one an on-chain empirical study published in the *Journal of Financial Economics*, the other NBER theoretical modeling—happen to provide a set of complementary evidence. The following strictly distinguishes three layers: "the empirical findings of the two papers / the authors' theoretical interpretation / the corresponding reading by this book's framework":

Dimension	Liu, Makarov & Schoar (2026) on-chain empirical findings	Uhlig (2022) theoretical interpretation	Correspondence in this book's framework
Failure of the external reserve anchor	UST's core redemption mechanism is not backed by off-chain reserves; in economic substance it resembles "infinite maturity convertible debt with a face value of $1 backed by LUNA"; LFG's roughly 80,000 BTC external reserve was insufficient and did not avail	——	§6.1 robustness of the external anchor: near-fully-closed multi-mapping recursion (limited external support that failed to work)
Subsidy-driven dynamics	The Anchor protocol attracted deposits with a yield of about 19.5%, an unsustainable dynamic in which newly issued tokens supported existing yields; the authors explicitly note it is "unlike a classical Ponzi scheme," declining to characterize it as a classical fraud	——	§5 Minsky directionality: a contemporary sample of pathological recursion in an environment centered on algorithmic redemption and lacking a prudential safety net

Dimension	Liu, Makarov & Schoar (2026) on-chain empirical findings	Uhlig (2022) theoretical interpretation	Correspondence in this book's framework
Observability paradox	On-chain observability accelerated the coordination of the run but did not create fairness—the complexity of the system let larger, more sophisticated investors exit faster with smaller losses, analogous to traditional bank runs (Iyer & Puri, 2012)	——	a boundary prompt to §4.1 (see below)
Gradual collapse	The collapse was not instantaneous but unfolded over three days	explains, through holders' heterogeneous exit thresholds, why the collapse was gradual rather than instantaneous	not forcibly mapped, recorded only as a characterization of process

Two implications. First, the two rows on the failure of the external reserve anchor and on subsidy-driven dynamics are consistent with the line of argument in §6.1 and §5: when the core redemption mechanism of a pricing system flows entirely back into a token internal to the system, and limited external reserves are insufficient to constitute an independent calibration source, it approaches the real-world limit of what this book calls "fully closed multi-mapping recursion"—under stress there was no adequate external calibration source to fall back on, so it destabilized within days (though not instantaneously; see the "gradual collapse" row) rather than drifting slowly. Second, the row on the observability paradox constitutes a meaningful **boundary extension** of, rather than a simple confirmation of, §4.1's reading that "the mapping chain is impenetrable": §4.1 assumes information loss comes mainly from "the mapping being hidden, the chain being opaque," whereas Terra is a counterexample in which on-chain data were highly public yet mapping breakdown still occurred, and the stampede was accelerated precisely by observability. This suggests that the de facto impenetrability of the mapping chain can arise, beyond "hiding," from "complexity itself"—even when the underlying data are highly public, differences in processing capacity still create substantive information asymmetry. It is recorded here only as a tension worth incorporating in later revisions.

§6.2 Maintenance Mechanisms of the Master Clock: Network Effects, Self-Reinforcing Feedback, and Independent Transmission

Chapter 6, §2 has already argued for four strategies of monopoly over the mapping function in AI, knowledge barriers, spatial barriers, temporal barriers, and narrative barriers. This chapter applies the same analytical framework to the global monetary system. The monopoly position of the dollar is not maintained by the same four strategies, but by a self-referential structure grounded in network effects, dollar -> Treasuries -> global benchmark -> further rise in the usage share of the dollar. The maintenance mechanisms differ, but the underlying structure is the same: the degeneration of mapping into self-reference.

The three empirical criteria test the transmission structure of the master clock from different dimensions: independence, whether the transmission channel is detached from spreads and trade; self-reinforcement, whether transmission forms a positive feedback loop; and lock-in, whether network effects obstruct substitution. The three bodies of evidence below respectively confirm the three dimensions.

Independence of the transmission mechanism. Acharya, Akinci, Miranda-Agrippino & Pesenti (2025), in a Liberty Street Economics analysis from the Federal Reserve Bank of New York, point out that the dollar is not only the dominant global currency; cross-border transmission through the dollar forms a monetary-policy spillover channel independent of interest-rate spreads and trade channels. Meisenzahl, Niepmann & Schmidt-Eisenlohr (2025), using high-frequency identification in the U.S. syndicated loan market, confirm empirically that a surprise 1% appreciation of the dollar against the euro during the syndication period widens the spread on a U.S. leveraged loan by up to 22 basis points, with spreads on higher-risk loans more sensitive to dollar appreciation (their 2021 study finds that a one-standard-deviation rise in the dollar index widens spreads by up to 15 basis points and reduces underpricing by up to 7 basis points). This means that the dollar's transmission effect is not merely the result of spreads or trade, but an independent channel of risk-preference transmission—the sensitivity of other financial conditions to dollar conditions exceeds the sensitivity of U.S. financial conditions to other economies (the latter requires separate testing with two-way transmission evidence). Miao Yanliang, Hao Yang & Fei Xuan (2021), on Chinese data, provide direct empirical verification of this independent transmission channel: historically, cross-border capital flows in China were driven mainly by exchange-arbitrage rather than interest-arbitrage capital, and the driving factor was the broad dollar index (DXY), not the bilateral RMB-dollar exchange rate or the China-U.S. interest-rate spread—"the exchange-rate effect is greater than the interest-rate effect, and the multilateral exchange-rate effect is greater than the bilateral exchange-rate effect" (Miao et al., 2021, author's translation). This is precisely the Chinese empirical validation of the master-clock concept: the dollar index, as a synthetic reading of global financial conditions, has transmission power beyond the spread channel and directly reshapes China's capital flows through exchange-rate expectation and risk sentiment.

The self-reinforcing nature of the feedback loop. Yildirim & Erdal (2026), in *The North American Journal of Economics and Finance*, use three kinds of vector autoregression (VAR) models to identify systematically, within their sample and specification, the central role of the dollar in the transmission of global interest-rate shocks. Counterfactual analysis reveals that there exists a self-reinforcing feedback loop between the dollar and global financial risk—the dollar is not only a transmission channel but also an amplifier. Threshold analysis further shows that among three sources of state dependence, uncertainty state, dollar state, and business-cycle state, dollar-state dependence dominates the remaining sources in global financial dynamics. Time-varying analysis shows that the transmission role of the dollar has continued to strengthen after the global financial crisis, especially in the post-pandemic period. This means that the master clock is not merely a static central

node; its influence is dynamically increasing.

Lock-in of network effects. The BIS Triennial Survey 2025 shows that in April 2025 the dollar was on one side of about 89.2% of all foreign-exchange trades (a measure summing to 200% across both sides), up from 88.5% in 2022. SWIFT (Society for Worldwide Interbank Financial Telecommunication) payment-message statistics show that in December 2025 the dollar accounted for 50.49% of global payments within its coverage (SWIFT, 2026); on the trade-invoicing dimension, the latest IMF study, covering 132 countries over 1990–2023, shows that the dollar remains dominant, with global invoicing shares broadly stable (Boz et al., 2025). Rogoff (2025), from the perspective of a former chief economist of the IMF, gives the systematic argument: "Thanks to network effects, international currency usage is a natural monopoly. So it should not be surprising that once a currency establishes itself in international transactions, the use of most other currencies falls by the wayside" (Rogoff, 2025). Rogoff (2025) uses the exchange between the Turkish lira and the Bulgarian lev as it stood then to illustrate the network advantage of the dollar as intermediary currency—an example belonging to the market structure before Bulgaria's adoption of the euro in January 2026, though the inertia of the intermediary currency it illustrates does not vanish because a single country joins the euro area. The formulation of Mark Williams, chief Asia economist at Capital Economics, at the 34th EuroFinance International Treasury Management Conference, is structurally identical but simpler: "You use the currency that is most readily available. That is the dollar — and because everyone uses it, it stays cheap and accessible. It is a self-reinforcing loop" (Williams, 2025). This does not come from any intrinsic value in the dollar itself, but from the self-referential structure of network effects—the degeneration of mapping into self-reference ($\pi_i(T) \to \pi_i(\{r_j\}_{j\neq i})$) at the institutional scale of the global monetary system.

§6.3 Conditions of Master-Clock Transition and Multi-Scale Diagnosis

Conditions of master-clock transition. The framework does not presuppose that the dollar will forever occupy the position of the master clock. Transition of the master clock, for example toward a multi-currency system, is possible. Its critical conditions are precisely the object of L3 recursive-closure diagnosis: when the deviation of the external anchor, U.S. sovereign credit, exceeds a critical threshold, and the network effect of substitute currencies exceeds a critical scale, recursion may slide from functionality into pathology. Abadi, Fernández-Villaverde & Sanches (2025), building a general-equilibrium model of international currency competition from micro foundations, provide formal conditions for master-clock transition: the model supports both a "unipolar equilibrium," one currency dominant, and a "multipolar equilibrium," multiple currencies circulating in parallel. Large economies enjoy a natural advantage in internationalizing their currencies—the probability is higher that foreign agents will transact with domestic agents of a large country. But when the dominant country lacks sufficient fiscal capacity to absorb global demand for liquid assets, a multipolar equilibrium emerges (Abadi et al., 2025). This is essentially the micro-founded version of the Triffin dilemma: maintaining the master clock requires continuous issuance of debt to satisfy global liquidity demand, but debt issuance itself erodes the external anchor of the master clock, sovereign credit. When the fiscal-capacity constraint tightens, the conditions for the transition from "unipolar -> multipolar" are in place—exactly the structural threshold that must be monitored by L3 recursive-closure diagnosis.

Multi-scale diagnosis of the state of the master clock. Rogoff (2025), from a long-historical perspective, provides the large-time-scale reference for master-clock transition: "the mantle of the world's dominant currency does not change hands very often. One to two centuries is the norm, and the transition is typically marked by the

co-existence of both the old dominant currency and the eventual new champion." From the Spanish silver 8-real piece of the sixteenth century, with silver-standard China in the same period forming a parallel Asian monetary pole, to the Dutch guilder in the seventeenth, to the pound sterling of the nineteenth and twentieth centuries, while China remained on silver until fiat reform in 1935, and then to the dollar from the mid-twentieth century onward, every shift of the master clock was accompanied by war, financial innovation, or both. Before Bretton Woods, the "master clock" never truly covered the whole globe; it was always a regional unipolarity paralleled by a silver-standard system, China, India, parts of Latin America, that formed the other pole.

Rogoff's judgment is that the era of dollar dominance "having lasted just over a century, may be regarded as late middle-aged" (Rogoff, 2025). This gives L3 recursive-closure diagnosis a dual time-scale reference: master-clock transition is not an annual event, but its early signals, deviation of the external anchor and expansion of substitute networks, can be identified on an annual time scale. IMF COFER data show that the dollar's share in global foreign-exchange reserves decreased to 56.77% in Q4 2025 from 56.93% in Q3 2025 (IMF, 2026). A quarterly change of 0.16 percentage points is not in itself a basis for judgment, but it flags the external-anchor dimension for continued monitoring. Pradhan et al. (2026), using the BIS International Debt Securities database, 1966–2024, provide an important correction to the narrative of "monotonic de-dollarization": the share of the dollar in IDS outstanding shows a "wave-like" rather than monotonic pattern, with three waves of dollarization, peaks in 1984 and 2000 and a third wave after the global financial crisis, after which "the share of the dollar in 2024 is close to its level in 2000" (Pradhan et al., 2026). Whether a decline in reserve share is structural slippage or a "wave reversal" is precisely what L3 diagnosis must determine—until the evidence suffices to distinguish the two, neither should be presumed.

§6.4 The Decoupling of Currency Privilege and Treasury Privilege

Du, Keerati & Schreger (2025), using SOFR after benchmark-rate reform to remeasure the convenience yield of the dollar and of U.S. Treasuries, find that the two have significantly decoupled after the global financial crisis. The convenience yield of the dollar remains positive and strengthens further when global risk rises, whereas the convenience yield of U.S. Treasuries has not only declined sharply, but has turned negative at medium and long maturities: "U.S. Treasury securities are no longer inherently more special than synthetic dollar bonds created by swapping foreign government bonds in G10 currencies" (Du et al., 2025). This decoupling shows that the network effect of the dollar as money, convenience in payment and funding, and the relative attractiveness of Treasuries as safe assets are separating. The "currency privilege" and the "Treasury privilege" of the master clock are no longer synchronized. Which of the two is changing faster must be judged from their respective time-series data (the reserve share and the payment share cited in this chapter are single-period observations and do not constitute a trend).

§6.5 Institutional and Narrative Consequences of the Master Clock

The influence of the master clock goes beyond pricing. Through intermediating links such as funding dependence, international-institutional rules, and policy conditionality, dollar dominance influences what other countries judge to be the "correct economic system." The policy framework labeled the Washington Consensus—especially the version later implemented: open capital accounts and financial liberalization—pre-installs a default concerning what counts as the proper economic system. Within this framework, "capital controls" automatically count as "backward." Even before argument begins, the framework classifies them as abnormal states

needing justification.

The Asian financial crisis of 1997 provides the case. The IMF prescribed the same medicine to Thailand, Indonesia, and Korea: fiscal tightening, higher interest rates, and open capital accounts. Malaysia's Mahathir refused and imposed capital controls; his policies were attacked from all quarters. The result was that the countries accepting the medicine suffered severe contraction; Malaysia, after imposing capital controls, went through a shorter contraction—its relative recovery performance depends on comparison dates and methods, and the scholarly assessment is contested. Stiglitz (2002), as former chief economist of the World Bank, offers a systematic critique: the failure of the IMF was not a technical misjudgment, but a structural one. Its analytical framework was preloaded by ideology and systematically unable to see realities inconsistent with the preloaded framework.

The institutional choices of different countries are, in essence, searches for different equilibria between the cost of control and the fragility of the financial system. The lesson of the Asian crisis is that both policy paths carried costs, and the frame "free market vs. planned economy" restricts the options to those two, thereby bypassing a more practical question: under either system, can financial products, at bearable information cost and within regulatory time, be traced to the underlying cash flows, the bearers of risk, and the resources that service repayment? This is not an ideological dispute, but a criterion of institutional design. Canada provides an instance of such design, see §3.3. This section focuses on the L3 layer—the institutional consequences in recipient countries of the structural stability of the master clock, the squeeze that the Washington Consensus exerts on the space of policy choice. The filtering mechanism of that ideology at the L2 layer—the visible-hidden subspace partition of openness vs. control, and the attribution shift in stress testing—is treated in a paired discussion in Chapter 20, §5.5.

§6.6 Diagnostic Boundary and Forward Guidance

The boundary between functional recursion and pathological recursion. The slide of the master clock from functional recursion to pathological recursion means that the external anchor of the global financial mapping network is being systematically suppressed. This is precisely the institutional-scale manifestation, in the global monetary system, of the Ponzi-finance phase in Minsky described in §5. When enough market participants cease independently to judge the true value of dollar assets and instead take "other people are still buying" as the entire reason for holding them, the self-referential mapping of the master clock has slid from functional recursion to pathological recursion. The boundary between the two is probabilistic. The four cases in Chapter 22 will provide ex post empirical reference points for the conditions of slippage. The task of L3 diagnosis is precisely to track these signals and identify the probabilistic critical conditions under which master-clock recursion slides from functional to pathological, rather than to assert that the master clock is eternal. What the framework can currently provide—the location and tracking of the above signals—belongs to the level of qualitative early warning. More refined quantitative indicators required to identify the critical condition of master-clock transition, such as the critical threshold at which the dollar's existing network destabilizes, exceed the current capacity of the Fiber Framework.

Recent work in 2025–2026 reveals two structural changes in the transmission mechanism of the global master clock. First, Auray et al. (2026) find that in 2025 the term structure of inflation expectations displayed a historically unprecedented significant negative correlation: for the first time the one-year inflation swap rate and the one-year-forward one-year inflation swap rate became significantly negatively correlated. The study finds

that the decoupling occurs mainly on trading days when tariff news dominates market pricing, and that the standard small-open-economy model they examine cannot explain the pattern. To replicate the observed negative correlation, one must introduce three conditions simultaneously: targeting year-on-year inflation, significant interest-rate inertia, and persistent tariffs. In fiber language, this means that the term structure of dollar inflation expectations—a component of the master clock's signal structure—is being rewritten by policy shocks, tariffs.

Second, the cross-country study of Curcio, D'Amico, Hasan & Vioto (2026) confirms the structural impact of digital finance on systemic risk: during market downturns of up to 10%, FinTech and BigTech have a greater impact on systemic risk than crypto-assets, while crypto-assets' more extreme declines, though smaller in magnitude, still lead to statistically significant increases in the systemic risk of banks and insurers; the relation between banks and technology companies shapes the channels of risk transmission; and "shock transmission from the US and EU to emerging markets systematically stronger than in the reverse direction." The transmission of the global master clock is not only one-way, U.S. -> world, but stratified. The transmission intensity from core countries to peripheral countries persistently exceeds reverse transmission. This is the concrete manifestation of the "two-waying of partial-input networks" described in §1.2: the influence of the master clock remains asymmetric, but channels of reverse transmission have already opened. (For a case-anatomy sample of an on-chain stablecoin run—containing the three diagnostic clues of a failed external reserve anchor, subsidy-driven dynamics, and the observability paradox—see the "Terra-Luna case box" at the end of §6.1.)

§7 Conclusion: The Duality of Financial Clocks

The financial system is a mapping system that repeatedly swings between two states: at one end, clocks anchored in the real economy; at the other end, self-referential clocks. The two states share the same underlying mechanism—price discovery, risk management, and intertemporal allocation. The difference lies in whether the mapping chain can be effectively penetrated, whether risk-bearing is matched with decision rights, and whether calibration results enter into actual constraints—chain length is only one dimension.

Securitization enabled millions of Americans to buy homes. Derivatives helped airlines manage fuel-price risk (Southwest Airlines, 2009). NASDAQ provided the financing platform for Apple, Microsoft, and Amazon. Under suitable informational conditions the price-discovery mechanism did in fact aggregate dispersed knowledge into effective signals. These are the social values of financial clocks when they point to the real—not rhetoric, but fact. But the diagnosis accumulated in this chapter is equally fact.

The nearly century-long evolution of the U.S. financial system displays the other side: the New Deal established calibration institutions -> forty years of stability -> deregulation dismantled calibration -> the S&L crisis -> still more deregulation -> LTCM -> still more deregulation -> 2008 -> Dodd-Frank reconstructed calibration. Minsky's proposition is verified in institutional history: maintaining calibration requires continuous work, while the return on that work is invisible in stable periods.

The core judgment of this chapter is that whether channels of calibration exist and are continuously maintained, together with whether the mapping chain can be penetrated, jointly determines whether financial clocks serve the real or degenerate into self-reference. Maintaining calibration channels requires continuous institutional investment—and the motivation to make that investment is weakest in periods of stability. This is the institutional unfolding of the Minsky proposition. The Fiber Framework's diagnosis hereby completes its first round of unfolding in the financial domain: the continuous work of maintaining calibration channels determines whether

the fiber is honestly marked or pseudo-shrunk—the next five chapters build layer by layer on this foundation.

The subsequent five chapters will unfold the diagnosis from different angles: Chapter 18 focuses on compensation systems and the revolving door, a China-US comparison of meta-audit at the institutional level; Chapter 19 traces the fundamental transformation of mapping-chain structure brought by the global 24-hour continuous market, a fiber diagnosis of speed, spatialization, and cross-market pricing; Chapter 20 reveals the ideological filter built into pricing models themselves, profit and value in the hidden subspace; Chapter 21 brings the macro diagnosis down into the micro condition of concrete financial products and practitioners; Chapter 22 unfolds the cross-period accounting and calibration of the MCAF through four financial-historical cases, three of which serve as principal-evidence cases with completed ex post calibration.

Terminology Correspondence Table

The correspondences between the financial phenomena introduced in this chapter and the core concepts of the Fiber Framework and the MCAF are as follows, providing a unified coordinate system for the next five chapters:

Financial phenomenon	Correspondence in the Fiber Framework	Correspondence in the MCAF
Underlying assets, mortgages, corporate loans	True state (T)	T—the object being mapped
Layer-by-layer packaging in MBS/CDO/CDO^2	Superposition of the mapping chain (L0), with the fiber inflating layer by layer as layers merge information	Securitized conversion of C (unlocked stock), turning originally illiquid loans into tradable securities
AAA assigned by rating agencies	Pseudo-calibration (L2)—a deterministic symbol conceals uncertainty	Inflation of ϕ (narrative load)—the symbolic value of AAA far exceeds its informational value
VaR models used in regulatory-capital calculation	Second-layer dynamic corrosion—the mapping function is rewritten through use	A (meta-audit capacity) is institutionally weakened
Hedge finance -> speculative finance -> Ponzi finance	Functional recursion sliding into pathological recursion (L3)	The three stages from accumulation to concealment to liquidation of cognitive debt
The dollar as global reserve currency	Master clock—other clocks define themselves against its readings	Very large Λ (media conductivity coefficient), and $\|L\|$ (legitimacy base) locked in by network effects
Positive feedback from market-cap weighting in passive investment	Accumulation of recursive components (new fund flows superimposed on natural weight changes, diluting independent components; see §1.2 on the two-waying of partial-input networks)	Positive-feedback amplification of Λ (media conductivity coefficient)

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$$\text{Incentive distortion} \xrightarrow{\text{ch18}} \text{Lengthening of the mapping chain}$$
$$\xrightarrow{\text{ch19}} \text{Institutional entropy increase} \xrightarrow{\text{ch20}} \text{Ideological masking}$$
$$\xrightarrow{\text{ch21}} \text{Drifting from the real into the virtual becomes a systemic tendency}$$
$$\xrightarrow{\text{feedback}} \text{Incentive distortion}$$

The four links are not isolated but mutually reinforcing and interlocking. Incentive distortion, convex compensation, short-term evaluation, the revolving door, drives practitioners to design more complex financial products, lengthening the mapping chain in order to extract more rent. The lengthening of the mapping chain creates regulatory blind zones, shadow banking, non-standard nesting, the speed predicament in algorithmic trading, and accelerates institutional entropy increase. Institutional entropy increase is then rationalized by ideological apparatuses, "the market is always right," "financial innovation improves efficiency." Ideological masking in turn protects incentive distortion, "high compensation is the natural outcome of the talent market." The four links together form a self-reinforcing loop. This transmission chain and the five-layer progressive structure complement each other: the five layers are a structural diagnostic tool, "what do we see at each layer?"; the transmission chain is a causal driving chain, "how does one layer drive the next?" In the later chapters, readers can track both dimensions simultaneously: which layer of the five each chapter deepens, and which link of the transmission chain each chapter occupies. The Chapter 22 calibration through the MCAF tests several key nodes of this transmission chain at the dynamic level, and the four financial cases there respectively show the consequences when the transmission chain is broken or interrupted at different nodes.